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A systematic literature review of urban consolidation centres:

Analysing service provision and its influence on distribution system performance from a logistics management perspective

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Abstract

Urban freight transport accounts for nearly a third of global freight-related greenhouse gas emissions and faces complex challenges driven by urbanisation and e-commerce growth. Urban consolidation centres (UCCs) offer a potential solution to lower the emissions by consolidating shipments to reduce vehicle trips and provide additional benefits, yet they face challenges related to operational success, financial sustainability, and viability. While most research focuses on transport aspects, investigations from a logistics management perspective considering factors like overall logistics costs and delivery services are limited. To address this gap, the purpose of this thesis is to review and synthesise existing research on UCCs from a logistics management perspective, focusing on services provided and their influence on distribution system performance.

A systematic literature review was conducted using a deductive approach guided by an analytical framework with four predefined service categories (delivery, warehousing, reverse logistics, and pre-retailing) and five performance attributes (time, cost, quality, productivity, and flexibility). Alongside the logistics management perspective, an underlying systems perspective was applied. A descriptive analysis and a thematic analysis were conducted. For the thematic analysis, a combined deductive and inductive category assignment was used, with the inductive elements added to deepen understanding of variation within the deductively derived categories. During the coding phase, a total of 20 service subcategories and 17 performance-related subcategories were formulated.

The analysis identified five key services provided by UCCs (consolidation, customised delivery time, long- and short-term storage, and recycling) which can be considered value-added as they deliver measurable benefits that shippers and receivers are likely to value. Delivery is the most frequently discussed category overall, while long-term storage is the most prominent subcategory. These services improve operational efficiency, vehicle fill rates, resource utilisation, and reduce urban delivery vehicles and costs. Cost is the most emphasised performance attribute, followed by productivity, with priorities differing across services. Benefits are context dependent, as factors such as additional distribution steps or limited adoption may offset gains. The findings highlight the need for a systems perspective to understand how service benefits and trade-offs interact across actors, distribution stages, and operational conditions. These findings carry both theoretical and managerial implications: they highlight the importance of a systems perspective for evaluating UCC performance and provide key considerations for designing and operating UCC services to improve overall performance and operational outcomes.

Keywords: Urban consolidation centre, Consolidation, Last-mile delivery, Urban freight distribution, City logistics, Systematic literature review.

Content

1. Introduction 1
1.1 Purpose and research questions 3
1.2 Scope and delimitation 3
1.3 Outline 4
2. Theoretical framework 5
2.1 Defining urban consolidation centre 5
2.2 Defining logistics management 5
2.3 Services 5
2.4 Performance attributes 7
2.5 Analytical framework 11
3. Methodology 12
3.1 Defining research questions 14
3.2 Defining criteria for inclusion and exclusion 14
3.3 Retrieving a sample of potentially relevant literature 14
3.4 Selecting relevant literature 15
3.5 Synthesising literature 16
3.6 Reporting results 19
4. Descriptive analysis 20
4.1 Characteristics of reviewed research 20
4.2 Characteristics of studied UCCs 21
5. Thematic analysis 24
5.1 Provided services 24
5.1.1 Delivery 24
5.1.2 Warehousing 25
5.1.3 Reverse logistics 26
5.1.4 Pre-retailing 26
5.1.5 Frequencies and distribution of services 26
5.2 Performance 30
5.2.1 Time 30
5.2.2 Cost 30
5.2.3 Quality 32

5.2.4 Productivity 32
5.2.5 Flexibility 33
5.2.6 Frequencies and distribution of performance 34
5.2.7 Frequencies and distribution of performance linked to services 37
5.3 The influence of services on performance 41
5.3.1 Improvements 41
5.3.2 Conditional improvements 42
5.3.3 Deterioration 43
5.3.4 Frequencies and distribution of the influence of services on performance 43
6. Discussion and conclusion 51
6.1 Key services that could be provided by a UCC 51
6.2 Emphasis of performance attributes in UCC contexts 52
6.3 How UCC services influence distribution system performance 55
6.4 Theoretical implications 59
6.5 Managerial implications 59
6.6 Limitations and further research 61
References 63

1. Introduction

Human activities, which release greenhouse gases (GHG) such as carbon dioxide, are the main cause of climate change and global warming (IPCC, 2018). In 2023, several disasters have occurred around the world that can be related to climate change, such as extreme and deadly heat waves, heavy rainfall, floods, intense wildfires, storms, and cyclones (Ripple et al., 2023). An activity that constitutes a significant contributor to GHG emissions and climate change is freight transport (Vleugel & Bal, 2018), which accounted for approximately 10% of the world's total GHG emissions in 2019 (ITF, 2023). In addition to the contribution to climate change, Vleugel and Bal (2018) emphasise that freight transports also have a negative influence on the quality of life and health through emissions of NO_X and particulate matter, noise, vibrations, congestion, and traffic accidents.

Although only a small fraction of the global freight transport activity (5%) takes part in urban areas, urban freight transport accounts for 28% of the total GHG emissions from freight transports (ITF, 2023), which can partly be explained by many short transports with small loads (ITF, 2021). Urban freight transport has, according to Björklund (2023), intensified due to an increased share of people living in urban areas. Mansouri et al. (2023) argue that urbanisation, together with the rapid rise of e-commerce, creates an increasingly complex and challenging logistical situation for deliveries in urban areas. In addition to the increased number of people living in urban areas and the growth of e-commerce, this complexity also stems from other factors, such as inadequate road infrastructure (CIVITAS WIKI, 2015), numerous actors (Björklund, 2023), and conflicting objectives (van Rooijen & Quak, 2014). The amount of urban freight transport is likely to keep growing, with ITF (2023) predicting that global urban freight transport activity will increase by a factor of 2.6 by 2050 compared to 2019 levels, following the trends of urbanisation and economic growth.

Freight transport is part of a distribution system designed to deliver products from the manufacturer to the customer through transport, storage, and goods handling (Oskarsson et al., 2021). Traditional distribution systems, which consist of a single direct transport from the manufacturer to the customer, often involve multiple transports with low vehicle fill rates (Lumsden et al., 2019). Oskarsson et al. (2021) describe the challenge of urban freight distribution similarly, where goods are ordered from various suppliers who employ different transport companies to make frequent deliveries with low vehicle fill rates to multiple customers within urban areas.

One approach to improving vehicle fill rates and reducing the number of required trips, thereby mitigating congestion and other negative effects of urban freight distribution, is the consolidation of goods (McDermott, 1975). Building on this practice, research on urban consolidation centres (UCCs) emerged in the early 1970s (Allen et al., 2012; McDermott, 1975). Unlike direct deliveries to receivers, UCCs consolidate shipments before final delivery, reducing the number of trips within urban areas. Figure 1 illustrates this difference, showing goods movement between actors, which may be handled by the shipper, the UCC, or an external carrier. The number of shippers, receivers, and flows varies depending on the UCC's geographic coverage and the types of goods it consolidates.

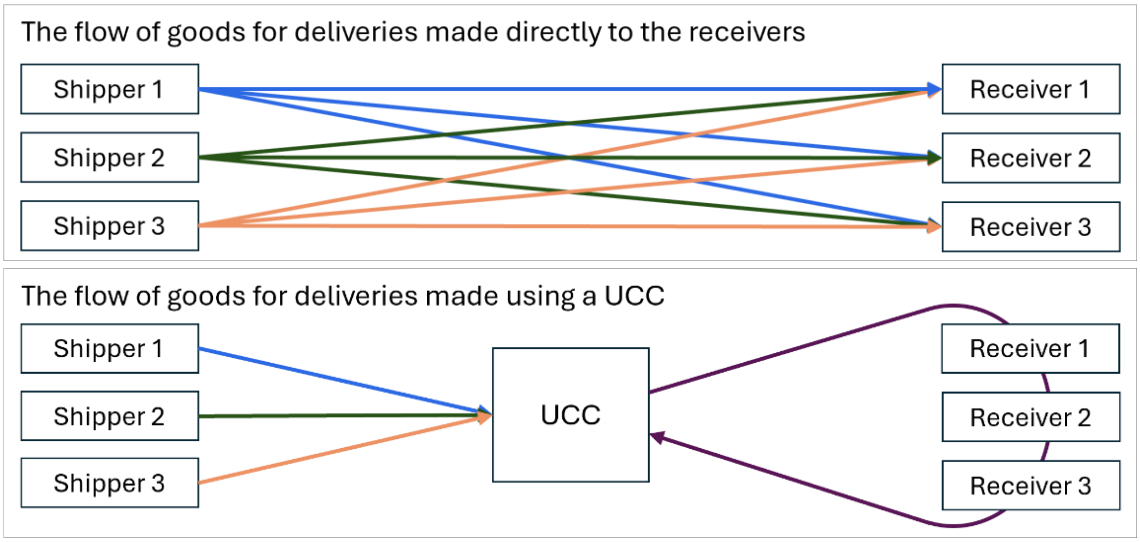


Figure 1: Comparison of the flow of goods before and after adding a UCC for urban freight transport

Johansson (2020) has identified several benefits that UCCs provide, including more reliable delivery times, reduced traffic congestion, lower risk of accidents, decreased emissions, and provide good publicity as an environmentally friendly initiative. However, Nylander et al. (2023) argue that despite increasing awareness of environmental impact, concerns for the environment and climate alone do not provide sufficient incentive to drive business change. Although carriers want to become more sustainable in their business operations, their opportunities are limited by the customer demands and expectations regarding lead time and price (Jazairy & von Haartman, 2019; Nylander et al., 2023).

A significant challenge for a UCC is ensuring financial sustainability (Allen et al., 2012; Björklund & Johansson, 2018). Introducing a UCC may meet resistance from carriers and retailers due to additional costs, but Browne et al. (2005) note potential for cost redistribution. Carriers may reduce time in congested areas, and retailers can save time by receiving consolidated rather than multiple deliveries. Nylander et al. (2023) highlight that cost savings can also come from better utilisation of resources such as space, staff, infrastructure, and equipment. Johansson (2020) further finds that UCCs can lower total costs for receivers through increased efficiency.

Still, prior research generally agrees that for a UCC to operate successfully, financial support may be necessary in various forms, including funding, subsidies, and incentives, until the UCC achieves financial sustainability (Allen et al., 2012; Mansouri et al., 2023; van Rooijen & Quak, 2014). Both Allen et al. (2012) and Tsiulin et al. (2017) propose that providing value-added services, that deliver advantages for which customers are willing to pay (Tsiulin et al., 2017), can enhance revenue and increase the UCC's attractiveness to potential users. However, providing such services will also increase the logistical complexity and costs of the operation (Allen et al., 2012). Johansson (2020) argues that, although financial viability is important, other factors also influence the viability of the UCC, i.e., its ability to operate successfully, such as providing services that not only meet customers' needs but also offer additional benefits. Slack and Lewis (2024) suggest that a business's ability to operate successfully can be assessed

by its operations' performance outcomes, which are evaluated through various attributes, such as time, cost, and quality. That said, there are several ways to measure, express, and interpret performance outcomes and attributes, and actors do not always value them in the same way (Slack & Lewis, 2024).

While previous literature reviews, such as those by Allen et al. (2012), Björklund and Johansson (2018), and Tsiulin et al. (2017), have addressed the importance of providing value-added services for the viability of UCCs, none has explored this topic in greater depth. Furthermore, Björklund and Johansson (2018) argue that research on UCCs has predominantly adopted a transport perspective, highlighting the lack of studies from a logistics management perspective that incorporate aspects such as total logistics costs and delivery services. This emphasis is also reflected in a more recent literature review by Gillström and Björklund (2024), which found that a significant portion of the included articles on UCCs were published in transport-oriented journals. However, while the transport perspective is widely adopted, it may overlook other operational considerations, such as total logistics costs, resource utilisation, and service design. A logistics management perspective is vital for understanding how UCCs can operate efficiently, remain financially sustainable, and meet customer expectations. This perspective also provides insights directly relevant to UCC operators, shippers, receivers, and policymakers seeking to optimise urban freight distribution and reduce environmental impact. Therefore, there is a need for a systematic literature review (SLR) that addresses both gaps: examining previous research on services provided by UCCs in relation to performance and approaching the topic from a logistics management perspective.

1.1 Purpose and research questions

The purpose of this thesis is to review and synthesise previous research on UCC from a logistics management perspective, focusing on the services provided and their influence on distribution system performance.

RQ 1: What key services could be provided by a UCC?

RQ 2: How are performance attributes emphasised in the context of UCCs?

RQ 3: How do the services provided by a UCC influence the performance of the distribution system?

1.2 Scope and delimitation

The review will be conducted from a logistics management perspective, concentrating exclusively on logistics-related services and performance. Additionally, the scope of this thesis is delimited to UCCs and does not include services provided by other actors within the distribution system, as illustrated in Figure 2. However, it is important to recognise that a UCC operates within this larger distribution system. Therefore, factors such as the range of services provided, as well as how these services are designed and delivered, may influence overall system performance. Accordingly, the needs and preferences of receivers and shippers regarding the UCC and its services are also taken into consideration.

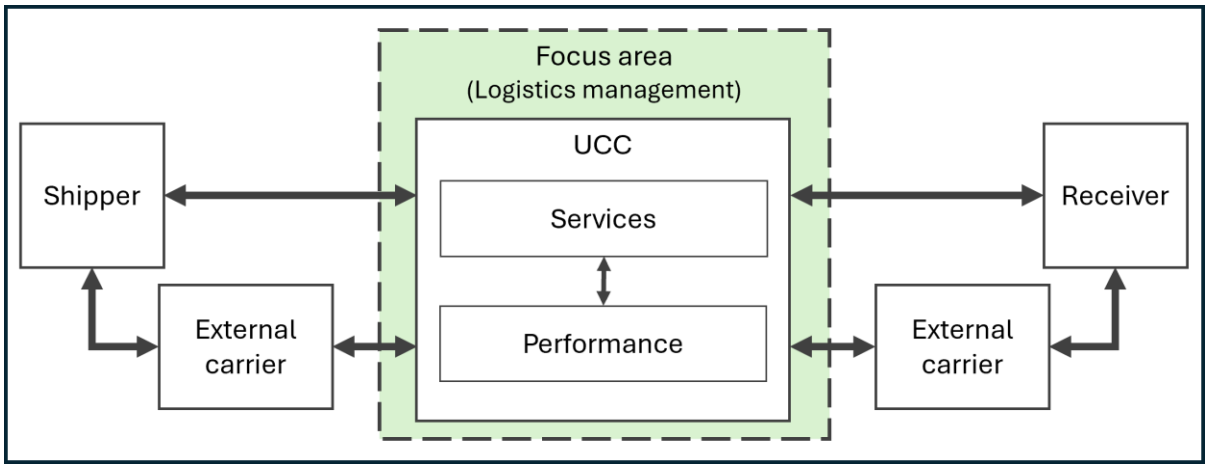


Figure 2: Scope of the thesis

1.3 Outline

The structure of the thesis is as follows.

Chapter 1 introduces the studied area, its challenges, and research gaps, leading to the thesis's purpose and research questions. It also presents the scope, delimitations, and outline of the thesis.

Chapter 2 reviews theoretical perspectives related to services and performance attributes, forming the basis for the analytical framework presented at the end of the chapter.

Chapter 3 explains the methodology, detailing each step of the systematic literature review and addressing related considerations and potential biases that arose during the process.

Chapter 4 outlines descriptive information about the included articles, including publication characteristics, research methods, UCC settings, and the types of goods handled.

Chapter 5 presents the thematic analysis of services and performance attributes, linking the findings to the research questions and distinguishing between existing and hypothetical UCCs.

Chapter 6 provides the discussion and conclusion, addressing the research questions, highlighting theoretical and managerial implications, and considering limitations and future research.

2. Theoretical framework

The theoretical framework defines key concepts such as UCCs and logistics management, and outlines the areas of services and performance, which together form the foundation of the thesis. Drawing on existing literature, four service categories and five performance attributes are identified, described, and summarised, establishing the analytical framework used in the study.

2.1 Defining urban consolidation centre

In this thesis, the definition of a UCC will be adopted from Browne et al. (2005):

> … a UCC is best described as a logistics facility that is situated in relatively close proximity to the geographic area that it serves be that a city centre, an entire town or a specific site (e.g. shopping centre), from which consolidated deliveries are carried out within that area. A range of other value-added logistics and retail services can also be provided at the UCC. Logistics companies with deliveries scheduled for the urban area or site are able to transfer their loads at the UCC and thereby avoid entering the congested area. The UCC operator sorts and consolidates the loads from a number of logistics companies and delivers them, often on environmentally friendly vehicles, to an agreed delivery pattern.
>
> (Browne et al., 2005, p. 4)

2.2 Defining logistics management

The definition of logistics management used in this thesis is provided by CSCMP (n.d.), which defines it as the planning, execution, and control of the flow and storage of goods, services, and related information (both forward and backward) between the point of origin and the point of consumption to meet customer needs in a cost-effective manner. In line with this definition, Oskarsson et al. (2024) argue that optimising a single part of the supply chain based solely on its own objectives is insufficient; instead, efficiency must be considered from a holistic perspective.

2.3 Services

According to Browne et al. (2005), UCCs can be designed in various ways depending on its geographic area and goals, resulting in a wide range of services. Its operation must consider the types of products handled and associated activities, including restrictions related to physical characteristics and regulations (Browne et al., 2005; Giampoldaki et al., 2023). For example, handling requirements may differ for products such as chemicals or food, for ambient versus chilled storage, or for items with special restrictions (Giampoldaki et al., 2023).

Based on literature, such as Allan et al., 2014, Browne et al., 2005, and Giampoldaki et al., 2023, four service categories are formulated for this thesis: delivery, warehousing, reverse logistics, and pre-retailing.

Delivery

The basic idea behind a UCC is to consolidate multiple deliveries into one before delivery to the receiver (Allen et al., 2014; Browne et al., 2005). In line with this, Giampoldaki et al. (2023) describe a UCC as a cross-docking facility, where goods are quickly transferred from inbound to outbound vehicles without intermediate storage (Lumsden et al., 2019). Browne et al. (2005) note that goods may arrive during off-hours, including nighttime, while outbound deliveries can follow a fixed schedule or be on-demand, such as within two hours. In addition to supplier deliveries, a UCC may handle store transfers, home deliveries, and delivery points (Browne et al., 2005). Its operations include coordinating shipments, transferring goods between transport modes, sorting and grouping items by receiver, and sequencing deliveries efficiently (Lumsden et al., 2019).

Warehousing

In addition to transport services, a UCC may provide warehousing, including storage and handling activities to manage goods effectively (Allen et al., 2014; Browne et al., 2005; Giampoldaki et al., 2023). Storage can include general or flexible areas as well as dedicated caged sections for specific retailers, depending on capacity and suitability for the goods, and typically addresses short-term needs such as seasonal or peak-period storage (Allen et al., 2014; Browne et al., 2005). According to Oskarsson et al. (2021), handling activities include arrival control, internal movement, storage, order picking, and preparation for delivery through packaging and labelling. UCCs may also check incoming goods for quantity, quality, security, or customs clearance when required (Allen et al., 2014; Browne et al., 2005; Giampoldaki et al., 2023).

Reverse logistics

Browne et al. (2005) emphasise that reducing transport impacts requires utilising the capacity of returning vehicles as much as optimising outbound deliveries. To use otherwise empty return trips, a UCC can provide reverse logistics services. Following Dowlatshahi (2000, p. 143), reverse logistics is defined as "a supply chain that has been redesigned to manage the flow of products or parts destined for remanufacturing, recycling, or disposal and to use resources effectively". This includes customer returns, used products, and materials suitable for remanufacturing, recycling, or disposal (Dowlatshahi, 2000). A UCC can support these flows by arranging return transport for unsold stock, packaging, waste, and damaged materials (Allen et al., 2014; Browne et al., 2005; Giampoldaki et al., 2023).

Pre-retailing

Pre-retailing includes services to prepare products for sale after delivery, such as unpacking, hanging goods after unpacking, preparing products for display, printing labels, labelling, relabelling, RFID/security tagging, kitting, assembling materials, and stocking shop shelves at the retailer (Allan et al., 2014; Browne et al., 2005; Giampoldaki et al., 2023).

Summary of service categories

A summary of the service categories is provided in Table 1. The summary offers a brief description of each service category along with the references used.

Table 1: Summary of service categories

Service categories	Description	References
Delivery	Provision of delivery services, including consolidation of deliveries, coordination of inbound and outbound transport flows, flexible timing for inbound deliveries, and fixed-schedule or on-demand options for outbound deliveries.	Allen et al. (2014); Browne et al. (2005); Giampoldaki et al. (2023); Lumsden et al. (2019)
Warehousing	Provision of warehousing services, including flexible and dedicated storage areas for short-term use, along with integrated handling and control activities such as arrival checks, order picking, quality and quantity control, and preparation for delivery.	Allen et al. (2014); Browne et al. (2005); Giampoldaki et al. (2023); Oskarsson et al. (2021)
Reverse logistics	Provision of reverse logistics services, including picking up unsold stock, packaging, waste, and damaged materials for remanufacturing, recycling, and disposal.	Allen et al. (2014); Browne et al. (2005); Dowlatshahi (2000); Giampoldaki et al. (2023)
Pre-retailing	Provision of pre-retailing services, including unpacking, hanging goods, preparing products for display, printing and applying labels, RFID/security tagging, kitting, assembling materials, and stocking retail shelves.	Allen et al. (2014); Browne et al. (2005); Giampoldaki et al. (2023)

2.4 Performance attributes

Various models in the literature describe categories or sets of performance measures for assessing operations (e.g., Jothimani & Sarmah, 2014; Slack & Lewis, 2024; Staudt et al., 2015). In the literature, different terms are used for these categories, such as "objectives", "attributes", and "dimensions". In this thesis, the term "performance attribute" will be used. Following Slack and Lewis (2024), each attribute includes smaller measures, or key performance indicators (KPIs), which provide partial insight into the attribute.

Slack and Lewis (2024) propose five generic objectives applicable to all operations: quality, speed, dependability, flexibility, and cost. Jothimani and Sarmah (2014) study the SCOR model for services, including reliability, responsiveness, flexibility, cost, and asset management efficiency. Staudt et al. (2015) suggest four performance dimensions: time, quality, cost, and productivity. Table 2 summarises the performance attributes of the models presented above, where similar attributes have been grouped, and based on this, five attributes are used in this thesis: time, cost, quality, productivity, and flexibility.

Table 2: Compilation of performance attributes in the literature

		Jothimani and Sarmah, 2014	Slack and Lewis, 2024	Staudt et al., 2015
Study area		Supply chain performance measurement	Operation management	Warehouse performance measurement
Performance attributes	Time	Responsiveness	Speed	Time
	Cost	Cost	Cost	Cost
	Quality	Reliability	Quality and dependability	Quality
	Productivity	Asset management efficiency		Productivity
	Flexibility	Flexibility	Flexibility	

Time

Time as a performance attribute relates to the duration of activities or processes and the speed of product delivery after an order (Jothimani & Sarmah, 2014; Slack & Lewis, 2024; Staudt et al., 2015). Oskarsson et al. (2021) note that reducing time can be a competitive tool, enabling cost savings and improved delivery service.

Common time-related KPIs include lead time, which describes the duration from start to completion of an activity or process (Faveto et al., 2024; Ghaouta et al., 2023; Jothimani & Sarmah, 2014; Oskarsson et al., 2021), cycle time (Faveto et al., 2024; Slack & Lewis, 2024), order lead time (Ghaouta et al., 2023; Slack & Lewis, 2024; Staudt et al., 2015), and delivery lead time (Ghaouta et al., 2023; Staudt et al., 2015). In warehouse management, KPIs can cover individual processes (e.g., picking time) or groups of processes (e.g., order lead time including picking, shipping, and delivery) (Staudt et al., 2015).

Cost

The cost attribute relates to expenses in running an operation (Jothimani & Sarmah, 2014; Slack & Lewis, 2024; Staudt et al., 2015). Slack and Lewis (2024) argue it is the most important attribute due to the focus on minimising business costs. Supporting this, Ghaouta et al. (2023) found cost was the most frequent attribute in warehouse performance studies. Oskarsson et al. (2021) emphasise considering total costs rather than individual departments, as reductions in one area may increase another. They propose a total cost model including warehousing, inventory, transportation, administration, and miscellaneous costs. Similarly, Staudt et al. (2023) argue that, in addition to assessing inventory levels monetarily, expenses such as holding costs, stockout penalties, depreciation, and insurance should be considered to obtain a complete measure of a warehouse's total costs.

Cost-related KPIs include inventory cost (Faveto et al., 2024; Ghaouta et al., 2023; Oskarsson et al., 2021; Staudt et al., 2015), storage cost (Faveto et al., 2024; Jothimani & Sarmah, 2014), distribution cost (Ghaouta et al., 2023; Staudt et al., 2015), and labour cost (Faveto et al., 2024; Staudt et al., 2015).

Quality

Quality performance can be assessed by whether a product or service fulfils its intended function, both in terms of specifications and whether these are met (Slack & Lewis, 2024). Staudt et al. (2023) distinguish five subgroups within quality: punctuality, completeness, correctness, breakage, and customer satisfaction. Punctuality, defined as orders being sent and/or received on the agreed date and time, aligns with Slack and Lewis's (2024) description of dependability. Oskarsson et al. (2021) similarly stress punctuality, or delivery reliability, noting its growing importance as companies reduce inventories and prioritise reliability over shorter lead times with wider delivery windows, driven by the need for smaller, more frequent deliveries. Staudt et al. (2023) describe correctness similarly to Oskarsson et al.'s (2021) delivery accuracy, delivering the right product, in the right quantity and quality, but extend it to include warehouse activities as well as delivery.

A common KPI related to quality is on-time delivery (Ghaouta et al., 2023; Oskarsson et al., 2021; Staudt et al., 2015). Other examples of quality-related KPIs include picking accuracy (Ghaouta et al., 2023; Jothimani & Sarmah, 2014), inventory accuracy (Ghaouta et al., 2023; Jothimani & Sarmah, 2014), and the perfect order (Ghaouta et al., 2023; Staudt et al., 2015).

Productivity

According to Lumsden et al. (2019), productivity involves managing personnel, financial resources, and physical assets (e.g., buildings, vehicles, equipment) to maximise output from a given input of resources. Productivity describes the output–input relationship, while utilisation concerns the ratio of utilised to available resources. A common utilisation measure is vehicle fill rate, which Björklund (2023) highlights as central for transports. Staudt et al. (2015) note that productivity measures, such as labour productivity, enable comparisons across contexts, for example between countries, since they relate work completed to working time regardless of variations in annual hours.

Examples of productivity-related KPIs are vehicle fill rate (Björklund, 2023; Ghaouta et al., 2023; Staudt et al., 2015), labour productivity (Slack & Lewis, 2024; Staudt et al., 2015), throughput (Faveto et al., 2024; Ghaouta et al., 2023; Staudt et al., 2015), and warehouse utilisation (Jothimani & Sarmah, 2014; Ghaouta et al., 2023; Staudt et al., 2015). Although Slack and Lewis (2024) placed productivity and utilisation-related KPIs under the cost attribute, this thesis follows Staudt et al. (2015) in treating productivity as an independent attribute, including utilisation-related KPIs.

Flexibility

Slack and Lewis (2024) identify two layers of flexibility: range and response flexibility, which concern the ability to adjust operations and the ease of modifying them in terms of time, cost, and smoothness, and four types of flexibility, product or service, mix, volume, and delivery, illustrated by variations in packaging, handling diverse product ranges, fluctuating demand, or faster delivery requests (Oskarsson et al., 2021; Slack & Lewis, 2024). While both Jothimani and Sarmah (2014) and Slack and Lewis (2024) suggest treating flexibility as an independent performance attribute, Staudt et al. (2015) view it as an indirect indicator due to measurement

complexity and links to other components, and Oskarsson et al. (2021) similarly describe it as a qualitative, "soft" attribute not measurable in the same way as others.

Examples of flexibility-related KPIs in the literature include capacity flexibility (Faveto et al., 2024; Ghaouta et al., 2023; Jothimani & Sarmah, 2014; Slack & Lewis, 2024), volume flexibility (Ghaouta et al., 2023; Jothimani & Sarmah, 2014), delivery flexibility (Ghaouta et al., 2023; Jothimani & Sarmah, 2014), and warehouse space buffer (Ghaouta et al., 2023; Jothimani & Sarmah, 2014).

Summary of performance attributes

A summary of the performance attributes is provided in Table 3. The summary offers a brief description of each performance attribute along with examples of KPIs and the references used.

Table 3: Summary of performance attributes

Performance attributes	Description	Examples of KPIs	References
Time	Time-related performance, focusing on the duration of activities and processes to enhance cost efficiency and delivery service.	Lead time, cycle time, order lead time and delivery lead time	Faveto et al. (2024); Ghaouta et al. (2023); Jothimani & Sarmah (2014); Oskarsson et al. (2021); Slack & Lewis (2024); Staudt et al. (2015)
Cost	Cost-related performance associated with the business's operations.	Inventory cost, storage cost, distribution cost, and labour cost	Faveto et al. (2024); Ghaouta et al. (2023); Jothimani & Sarmah (2014); Oskarsson et al. (2021); Slack & Lewis (2024); Staudt et al. (2015)
Quality	Fulfilment of product or service specifications and intended function, including punctuality, correctness, and customer satisfaction.	On time delivery, picking accuracy, inventory accuracy, and perfect order	Ghaouta et al. (2023); Jothimani & Sarmah (2014); Oskarsson et al. (2021); Slack & Lewis (2024); Staudt et al. (2015)
Productivity	Maximising the output of products and services from a given input of resources as well as the utilisation rate of available resources.	Vehicle fill rate, labour productivity, throughput, and warehouse utilisation	Björklund (2023); Faveto et al. (2024); Ghaouta et al. (2023); Jothimani & Sarmah (2014); Lumsden et al. (2019); Slack & Lewis (2024); Staudt et al. (2015)
Flexibility	Adaptability of operations to swiftly modify product or service offerings, ensuring fast, cheap, and smooth adjustments to meet changing customer needs.	Capacity flexibility, volume flexibility, delivery flexibility, and warehouse space buffer	Faveto et al. (2024); Ghaouta et al. (2023); Jothimani & Sarmah (2014); Oskarsson et al. (2021); Slack & Lewis (2024); Staudt et al. (2015)

2.5 Analytical framework

Figure 3 presents the analytical framework developed for this thesis. It builds on the thesis scope outlined in Figure 2. The framework incorporates the four categories of services described in Table 1, as well as the five performance attributes defined in Table 3. Together, these elements establish a structured basis for analysing the services provided by UCCs and their influence on the performance of the distribution system.

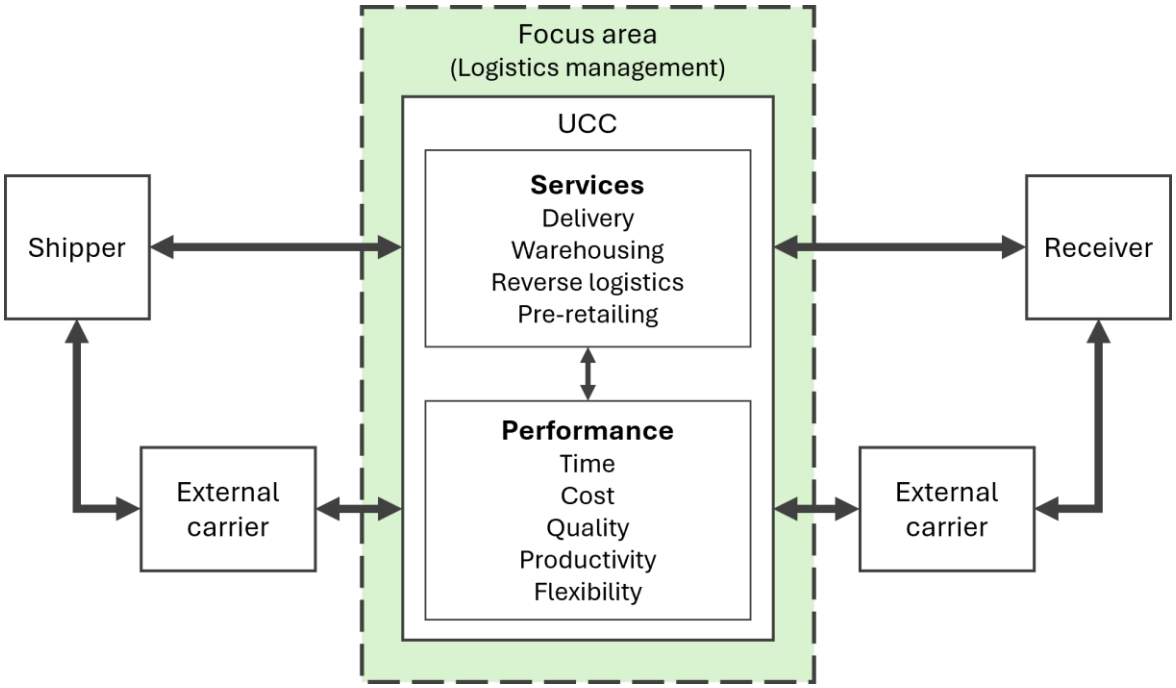


Figure 3: Analytical framework

3. Methodology

Säfsten and Gustavsson (2019) emphasise that the choice of method for a study should be based on the purpose and research questions. For this thesis, SLR was chosen as method, as it was considered appropriate to fulfil the thesis's purpose of reviewing and synthesising previous research. The SLR was conducted according to the six-step guidelines outlined by Durach et al. (2017): (1) defining research questions, (2) defining criteria for inclusion and exclusion, (3) retrieving a sample of potentially relevant literature, (4) selecting relevant literature, (5) synthesising literature, and (6) reporting the results. A review and synthesis of previous research makes it possible to create an overall summary of the evidence, what is known, and to identify similarities, contradictions, and potential gaps (Booth et al., 2017).

Throughout the process, three key considerations identified by Booth et al. (2017) as crucial for a systematic literature review were taken into account: clarity, validity, and auditability. To enhance clarity and auditability, the research process will be presented, starting with an overall illustration of the six steps of the SLR process (see Figure 4), detailing key information for each step, including criteria for inclusion and exclusion, the search string, and the number of articles in the sample. This will be followed by subheadings for each of the six steps, where a more detailed description will be provided. The considerations made during the process will also be discussed in relation to various biases and other validity-related issues.

During the SLR process, a large number of articles were managed. To facilitate handling the sample articles, Rayyan, an AI-assisted tool for systematic literature review (Ouzzani et al., 2016), was used. During the article search in step 3, files were downloaded from Scopus and Web of Science, containing information about the articles, such as the title, author, year, volume, issue, DOI, and abstract. These files were uploaded to a review workspace created for this thesis on Rayyan, which then served as an article register throughout the rest of the process. The specific use of the tool is described in each of the steps below.

In the writing of the thesis, ChatGPT was employed to assist with grammar, language clarity, and suggesting alternative wording. Prompts clearly specified that no new information should be added, and all suggestions were carefully reviewed to ensure that the original meaning of the text remained unchanged.

Step 1: Defining research questions

Purpose	The purpose of this thesis work is to review and synthesise previous research on UCC from a logistics management perspective, focusing on the services provided and their influence on distribution system performance.
RQ 1	What key services could be provided by a UCC?
RQ 2	How are performance attributes emphasised in the context of UCCs?
RQ 3	How do the services provided by a UCC influence the performance of the distribution system?

Step 2: Defining criteria for inclusion and exclusion

Initial criteria set	**Inclusion**	**Exclusion**
Databases	Scopus and Web of Science	
Search fields	Article title, abstract, keywords, and Topic	
Source type	Peer-reviewed journal articles	
Language	English	
Method used		Literature reviews
Advanced criteria set		
Relevance	Articles contributing to fulfilling the thesis's purpose and addressing the research questions, i.e. addressing services provided by a UCC and/or how the performance of the distribution system is influenced	

Step 3: Retrieving a sample of potentially relevant literature

Search string Formulated as: UCC AND (Service OR performance)	("Urban consolidation cent*" OR "City consolidation cent*" OR "Freight consolidation cent*" OR "Urban delivery cent*" OR "City delivery cent*" OR "Urban distribution cent*" OR "City distribution cent*" OR "Urban freight cent*" OR "City freight cent*" OR "Urban transhipment cent*" OR "City transhipment cent*" OR "Urban distribution depot*" OR "City distribution depot*" OR "Urban transhipment depot*" OR "City transhipment depot*" OR "Urban consolidation platform*" OR "City consolidation platform*" OR "Freight consolidation platform*" OR "Urban freight platform*" OR "City freight platform*" OR "Urban freight terminal*" OR "City freight terminal*" OR "Urban terminal*" OR "City terminal*" OR "City hub" OR "Consolidation terminal*" OR "Logistic* hub" OR "Pick-up drop-off location*" OR "Cooperative delivery system*") AND (("Servic*" OR "Activit*" OR "Function*" OR "Task*" OR "Operat*" OR "Process*") OR ("Perform*" OR "Outcome*" OR "Achiev*" OR "Indicat*"))
Initial sample (after applying initial criteria set)	390 articles (Scopus: 365, Web of Science: 225, duplicates removed: 200)

Step 4: Selecting relevant literature

Final sample (after applying advanced criteria set)	17 articles

Step 5: Synthesising literature

For the descriptive analysis, general information about the articles in the sample was extracted using a predefined template.

In the thematic analysis, 220 unique quotes were extracted from the articles and coded according to the service categories and performance attributes, along with inductively derived subcategories (20 for services and 17 for performance attributes), the status of the UCC (existing or hypothetical), and whether the quote described an improvement, conditional improvement, neutral effect, or deterioration.

Step 6: Reporting the results

The results of the descriptive and thematic analyses were synthesised and presented in the following sections: the descriptive analysis in Chapter 4, the thematic analysis in Chapter 5, and finally discussed and concluded in Chapter 6.

Figure 4: Overview of the SLR process

3.1 Defining research questions

In the first stage of an SLR, the review's purpose and research questions are defined (Durach et al., 2017). The purpose of this thesis is to review and synthesise previous research on UCC from a logistics management perspective, focusing on the services provided and their influence on distribution system performance.

RQ 1: What key services could be provided by a UCC?
RQ 2: How are performance attributes emphasised in the context of UCCs?
RQ 3: How do the services provided by a UCC influence the performance of the distribution system?

In addition to defining the purpose and research questions, the first step also includes specifying the unit of analysis and study context, as well as developing a theoretical framework (Durach et al., 2017). This thesis is limited to the UCC, which constitutes the unit of analysis. Alongside the logistics management perspective guiding the review, a systems perspective was applied, reflected in the third research question, since a UCC is part of a larger distribution system and its services affect both individual actors and the whole system.

The theoretical framework covers definitions fundamental to the SLR, categories of potential UCC services, and logistics-related performance attributes. From this, an analytical framework with four service categories and five performance attributes was developed to structure the thematic analysis. As this framework was established before the analysis and not derived from the material, the approach is described as deductive (Mayring, 2014).

3.2 Defining criteria for inclusion and exclusion

Two sets of inclusion and exclusion criteria were defined: an initial set applied to the sample retrieved in step 3 and an advanced set used to assess the remaining articles in step 4. The criteria were inspired by Björklund and Johansson (2018), Durach et al. (2017), and Jazairy et al. (2023).

For the initial criteria, articles were retrieved from Scopus and Web of Science. The search string was applied to article title, abstract, and keywords (Scopus) and topic (Web of Science). Only peer-reviewed journal articles in English were included, while studies based solely on literature reviews were excluded, as the aim was to synthesise empirical findings rather than summarise previous reviews. No restrictions were placed on publication year.

The advanced criteria evaluated article relevance. To be included in the final sample, articles had to contribute to fulfilling the thesis purpose and research questions by addressing UCC services and/or how distribution system performance is influenced.

3.3 Retrieving a sample of potentially relevant literature

For the article search, three keywords were derived from the thesis purpose and research questions: UCC, service, and performance. The search string used was: UCC AND (services OR performance). Related search terms for each keyword are listed in Table 4. Various

combinations were tested to identify the most suitable one. To capture different word endings and spelling variations, asterisks were used to truncate terms, as recommended by Säfsten and Gustavsson (2019).

Table 4: Keywords and search terms

Keyword	Search terms
UCC	Urban consolidation cent*; City consolidation cent*; Freight consolidation cent*; Urban delivery cent*; City delivery cent*; Urban distribution cent*; City distribution cent*; Urban freight cent*; City freight cent*; Urban transhipment cent*; City transhipment cent*; Urban distribution depot*; City distribution depot*; Urban transhipment depot*; City transhipment depot*; Urban consolidation platform*; City consolidation platform*; Freight consolidation platform*; Urban freight platform*; City freight platform*; Urban freight terminal*; City freight terminal*; Urban terminal*; City terminal*; City hub; Consolidation terminal*; Logistic* hub; Pick-up drop-off location*; Cooperative delivery system*
Service	Servic*; Activit*; Function*; Task*; Operat*; Process*
Performance	Perform*; Outcome*; Achiev*; Indicat*

For the keywords "services" and "performance," search terms that were too specific (e.g., "off-hour deliver*") or too broad (e.g., "result*") were removed. For "UCC", a more inclusive approach was used. Browne et al. (2005) list many terms for UCC, some of which describe unrelated arrangements. To avoid excluding relevant articles, broad terms (e.g., "logistic* hub") were included in the search string. In total, 29 UCC-related search terms identified in previous literature (e.g., Browne et al., 2005; Björklund & Johansson, 2018) were used.

During search string development, discussions with the supervisor helped ensure completeness and reduce retrieval bias, which can occur if the search is insufficient and produce a sample not representative of the literature (Durach et al., 2017).

The search was conducted on June 21, 2024. After applying the initial criteria, 590 articles remained (365 from Scopus, 225 from Web of Science). Duplicates were identified using Rayyan based on abstract similarity and manually reviewed within the tool, resulting in 200 duplicates removed and leaving an initial sample of 390 articles.

3.4 Selecting relevant literature

The initial sample of articles was further evaluated using the advanced criteria set to determine whether each could contribute to fulfilling the thesis's purpose and research questions. The evaluation occurred in two phases: first, titles, keywords, and abstracts were reviewed to remove articles from unrelated fields (e.g., healthcare and biology); second, when relevance could not be determined, the full text was examined to ensure the UCC studied aligned with this thesis's definition and addressed services provided and/or distribution system performance. After this process, 17 articles were included in the final sample.

Although the sample of 17 articles is relatively small, this does not necessarily indicate a lack of relevance in studies on the concept of UCCs. Rather, it may reflect the limited amount of academic research in the field. Van Duin et al. (2010) noted that much of the existing material on UCCs originates from engineering companies or local authorities rather than from scientific publications, and Giampoldaki et al. (2021) emphasised that, despite the growing popularity of UCCs, there is still little relevant research published in academic journals. Therefore, even a small, focused sample can provide meaningful insights and help address the existing research gap.

The Rayyan tool supported the inspection process, allowing titles, keywords, abstracts, and, in some cases, full texts to be reviewed. Articles could be categorised as included or excluded, with tags and labels marking reasons for exclusion, such as irrelevant research area, misalignment with this thesis's UCC definition, or other reasons. This structured approach enabled revisiting excluded articles, tracking inclusions, and monitoring pending decisions.

Risks of bias were acknowledged, including the potential for incorrect designed criteria or subjective evaluations. Although multiple independent reviewers are recommended to manage such risks (Durach et al., 2017), this thesis was conducted by a single researcher. To reduce the risk of bias, discussions were held with the supervisor regarding the design of the advanced criteria and inspection approach. The supervisor provided guidance on the strategy used for evaluating articles, and for a few cases where inclusion was difficult to determine, advice was sought on how to apply the criteria consistently.

A particular challenge was assessing the relevance of articles addressing UCC sustainability performance (e.g., vehicle fill rate and transport distances). These were excluded because the relevant performance indicators constituted only a small part of the studies.

3.5 Synthesising literature

A descriptive and a thematic analysis were conducted on the 17 articles in the final sample. The descriptive analysis followed Tranfield et al. (2003) using a simple set of categories and an extraction form. Using a predefined Excel template, general information was extracted, including the author, year, journal, method, whether the studied UCC exists or is hypothetical, the country in which the UCC is located, and the types of goods it handles.

For the thematic analysis, a combined method of deductive and inductive category assignment was employed. This method follows the logic presented by Mayring (2014), in which the analysis begins with a deductive phase followed by an inductive phase. The deductive phase involved developing a predefined category system and systematically extracting content from the articles. By subsequently incorporating inductive elements into an otherwise deductive method, the analysis allows for a deeper understanding of the variation within the deductively derived categories. Figure 5 below illustrates the 11 steps of the process, synthesising both methods into the approach applied in this study.

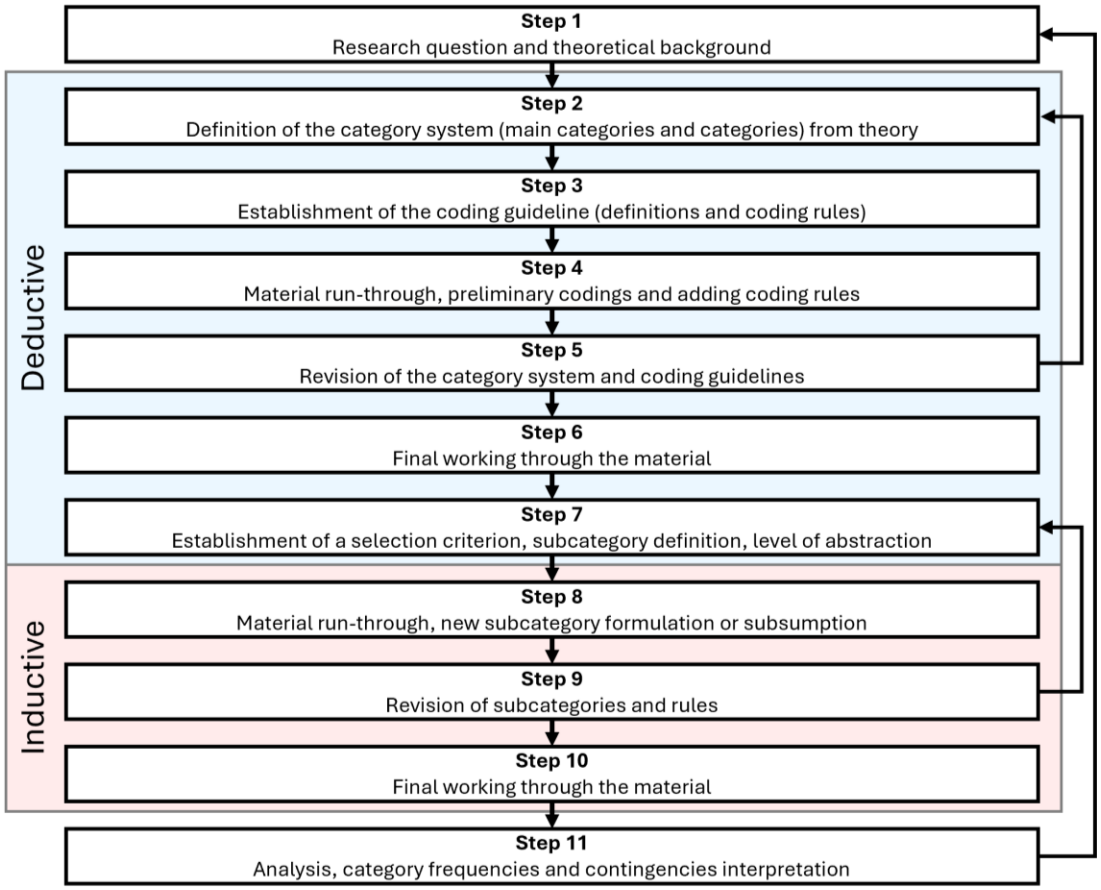


Figure 5: Overview of the thematic analysis process (synthesised and adapted from Mayring, 2014)

The thematic analysis began with formulating research questions and establishing a theoretical and analytical framework. The framework, derived from the literature, comprised four service categories and five performance attributes, grouped under the main categories of "services" and "performance attributes". The definitions provided in the framework guided the coding process and ensured consistency across articles.

Before coding, an Excel table was created with columns for the article, quote, and each main category and subcategory. Coding rules were formulated to guide the process, though not included in the table. For example, “distance travelled” was categorised under *time*, *cost*, or *productivity* depending on context, or excluded if unrelated to performance attributes. Similarly, profits and margins from a purely economic perspective (revenue minus costs) were not included under the *cost* attribute.

Quotes extracted from the articles were entered into the Excel sheet and marked against all relevant categories and attributes. If a quote matched multiple categories or attributes, it was marked against all relevant categories and attributes in each case. When additional coding rules were needed, such as for profits and margins, the guideline was revised and the coding process repeated to ensure consistency. This iterative procedure continued until all quotes were coded systematically.

This marked the start of the second, inductive phase of the thematic analysis. Selection criteria were established to define subcategories and their level of abstraction. In the first coding round, the criteria were kept broad, then refined through several test rounds. For service categories, subcategories were defined as specific services, such as off-hour delivery, recycling, and long-term storage. In total, 20 service subcategories were formulated: 11 for delivery, three for warehousing, three for reverse logistics, and three for pre-retailing.

For performance attributes, subcategories were tailored to each attribute using their definitions as a guide. For instance, *time* subcategories were based on activities and processes, *quality* on different quality parameters, and *cost* on cost components and system integration. In total, 17 performance-related subcategories were formulated: three for time, four for cost, three for quality, four for productivity, and three for flexibility.

Quotes of a more general nature that did not fit the formulation of the subcategories were placed in a temporary subcategory called "general" to facilitate the coding process. These quotes were still included, as they supported the category and attribute they corresponded to.

To understand how UCC services influence distribution system performance, subcategories were added within performance attributes to code quotes as reflecting improvement or deterioration. Some cost-related quotes were coded as conditional improvements, indicating improvement was possible only under certain conditions, such as reaching a specific throughput level. Quotes that did not clearly indicate improvement, conditional improvement, or deterioration were coded as neutral. This iterative process continued until all quotes were coded consistently. One cost-related quote addressed both improvement and deterioration and was therefore entered twice, once as improvement and once as deterioration.

Each quote was also coded as referring to an existing or hypothetical UCC. Hypothetical UCCs include centres planned or proposed for future implementation, or conceptual centres exploring a UCC's potential where none exist. This enabled comparison between the two types. Six articles examined only existing UCCs, nine only hypothetical, and two included both. In the latter cases, context determined which type a quote referred to. One quote addressed both types and was therefore entered twice, once for existing UCCs and once for hypothetical UCCs.

In total, 222 quotes were entered into the Excel sheet, of which 220 were unique. Of these, 84 referred to existing UCCs (across eight articles), while 138 referred to hypothetical UCCs (across 11 articles).

For the final stage of the thematic analysis, both qualitative and quantitative approaches were applied, with the qualitative analysis serving as the primary method and the quantitative analysis providing support.

In the qualitative analysis, quotes within each service subcategory were first synthesised to show how the services were described. Next, quotes within each performance attribute subcategory were then synthesised to clarify performance descriptions. Links between services and performance attributes were subsequently analysed to identify which attributes were

discussed in connection with each service and how services influenced distribution system performance. Links were established between service subcategories and the five performance attributes, and between performance subcategories and the four service categories, to examine variations in performance descriptions. No links were made between service subcategories and performance subcategories.

In the quantitative analysis, frequencies were calculated to show how often each category, attribute, and subcategory of services and performance attributes appeared in quotes and across articles. Analysing both quote- and article-level data provided a nuanced understanding, preventing overrepresentation of subcategories frequently mentioned in a single article or underrepresentation of those cited rarely across many articles. This approach helped identify key services, addressing the first research question, while combining quantitative and qualitative analysis enabled a comprehensive examination of how performance attributes are emphasised generally and in relation to specific services, addressing the second research question.

The quantitative analysis also assessed links between each service category and performance attributes by counting quotes describing performance as improvement, conditional improvement, or deterioration. This complemented the qualitative findings for the third research question on how services influence distribution system performance. The analysis also distinguished whether quotes referred to existing or hypothetical UCCs to explore potential differences.

As in earlier SLR stages, bias may occur during coding and synthesis, typically reduced by multiple researchers (Durach et al., 2017). To mitigate this risk in a single-researcher review, several steps were taken. The supervisor was consulted on the Excel table and template design to ensure relevant data were captured. A small sample of quotes was initially coded and discussed to confirm alignment with category and subcategory definitions. Based on feedback, overlapping categories were merged, some subcategories renamed, and definitions clarified, improving coding reliability. Throughout the iterative process, preliminary results and coding decisions were reviewed with the supervisor to identify inconsistencies and confirm accurate interpretation.

3.6 Reporting results

Descriptive findings are shown with bar graphs to visualise distributions across data categories, sometimes including additional details such as data types by method and the number of existing versus hypothetical UCCs handling perishable and non-perishable goods. Thematic analysis findings are mainly presented as a qualitative cumulative summary, with quantitative results displayed in diagrams and tables. The quantitative results complement the qualitative findings by highlighting which categories and subcategories are most prominent, providing insights not always evident from the qualitative analysis alone.

4. Descriptive analysis

The descriptive analysis provides an overview of the reviewed articles, covering publication details, research methods, and the location, status, and goods handled by the studied UCCs. The chapter ends with a compilation summarising the number and types of UCCs studied and the perspectives considered.

4.1 Characteristics of reviewed research

The articles were published from 2015 onwards. There has generally been a steady but low output of articles, with some exceptions (see Figure 6). The number of published articles peaked in 2017 with seven, before returning to the general pattern with one article in 2018 and two in 2019. No articles were published in 2020 or 2023.

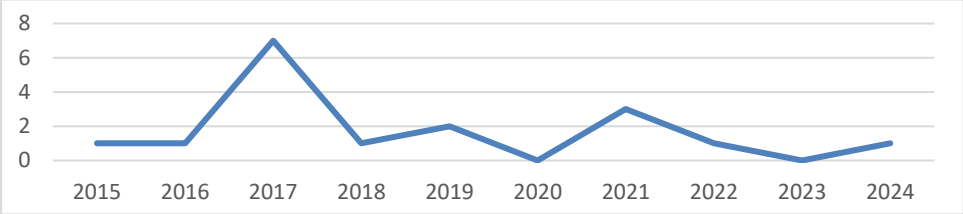


Figure 6: Distribution of publications over years

The articles are published in eleven different international journals (see Figure 7). Most journals contain only one article, while three journals contain more (2, 3, and 4 articles, respectively). Only two of the articles are published in logistics-oriented journals (*Supply Chain Forum* and *International Journal of Physical Distribution and Logistics Management*), while a substantial majority of the articles, 13 out of 17, are published in transport-oriented journals, such as *Research in Transportation Economics*, *Case Studies on Transport Policy*, and *Transport*. These findings are in line with the observations made in the literature reviews by Björklund and Johansson (2018) and Gillström and Björklund (2024).

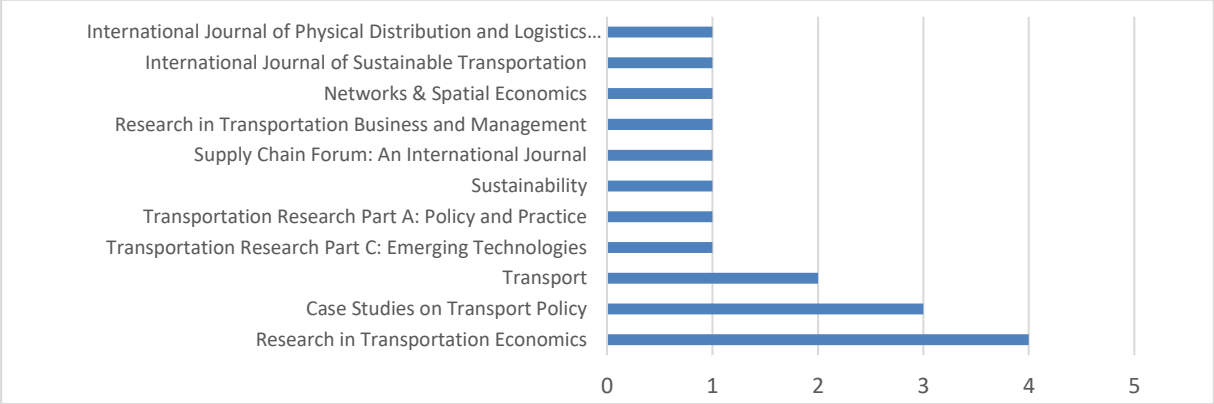


Figure 7: Distribution of journals

The research methods applied in the articles reviewed have been categorised in this thesis into three main groups: case study, survey, and modelling. As shown in Figure 8, the case study method is the most commonly applied across the articles. Among those that employ modelling, three distinct approaches are identified: mathematical modelling, analytical modelling, and agent-based simulation.

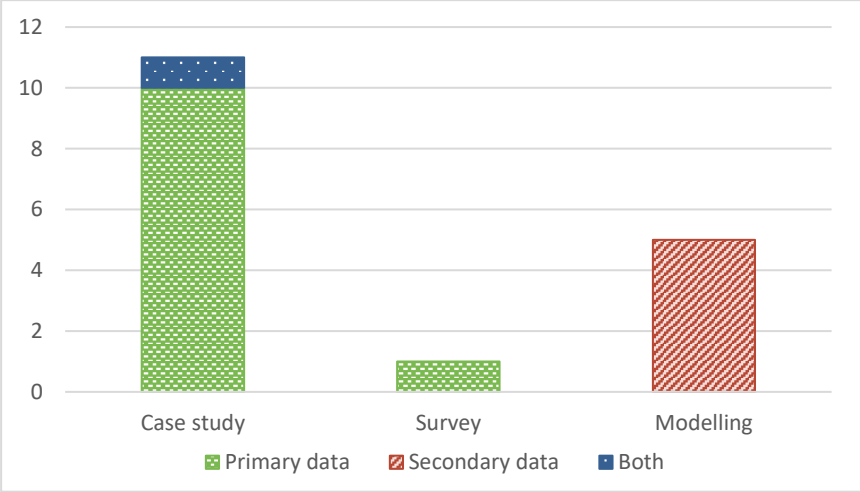


Figure 8: Research methods applied

Figure 8 also presents the distribution of primary and secondary data across the different methodological groups. Notably, the figure highlights the predominance of primary data, which was most often collected through interviews and questionnaires. This contrasts with previous literature reviews in the field of UCCs, where secondary data has typically been the main source in the analysed articles (Björklund & Johansson, 2018; Gillström & Björklund, 2024). All articles using modelling methods relied exclusively on secondary data, whereas one case study article used a combination of primary and secondary data.

4.2 Characteristics of studied UCCs

A total of 24 UCCs were studied across the reviewed articles, although these do not represent 24 unique initiatives, as the same UCC may have been examined in multiple sources. Figure 9 provides an overview of the regional profiles of these UCCs, indicating the country and continent in which each is located, as well as their implementation status. The data reveal a strong geographic concentration in Europe, where nearly all of the studied UCCs are situated. Only one UCC is located in Australia and one in North America, with no cases examined in regions such as Asia, Africa, or South America. The countries with the highest number of studied UCCs are the UK (6 centres), Italy (5 centres), and Sweden (4 centres).

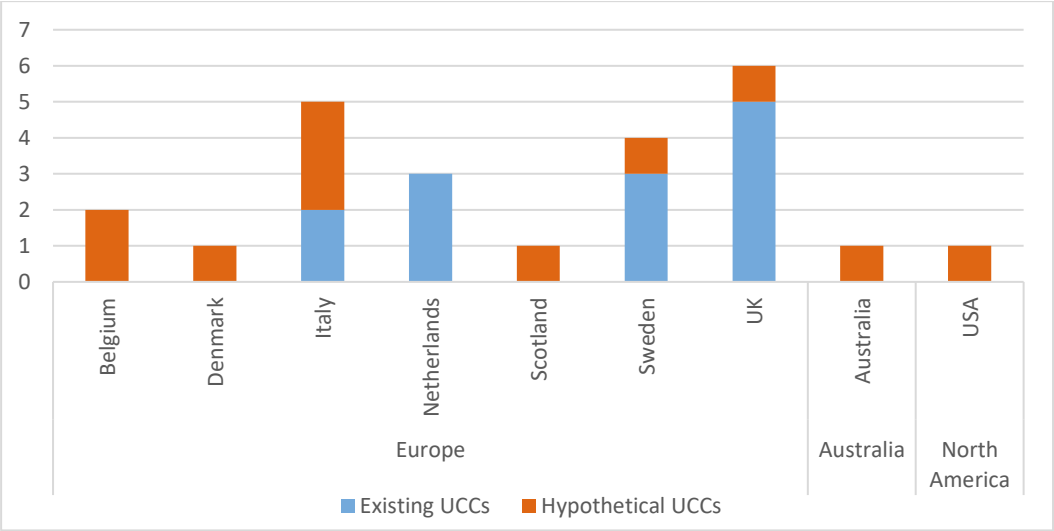


Figure 9: Regional profiles and implementation status of the studied UCCs

Regarding implementation status, the review includes slightly more existing UCCs than hypothetical ones, with 13 existing and 11 hypothetical cases studied. Notably, Figure 9 shows that both UCCs located in Australia and North America are hypothetical, meaning that all existing UCCs studied in the articles are in Europe. Among the existing UCCs, the highest numbers are found in the UK (5 centres), Sweden (3 centres), and the Netherlands (3 centres).

As shown in Figure 10, half of the studied UCCs, comprising six existing and six hypothetical centres, handle only non-perishable goods, such as fashion items, office supplies, and other non-perishable products. At the same time, two hypothetical UCCs focus exclusively on perishable goods, while two existing and one hypothetical UCC handle both non-perishable and perishable goods. For six existing UCCs and one hypothetical UCC, the type of goods handled is not specified in the reviewed material.

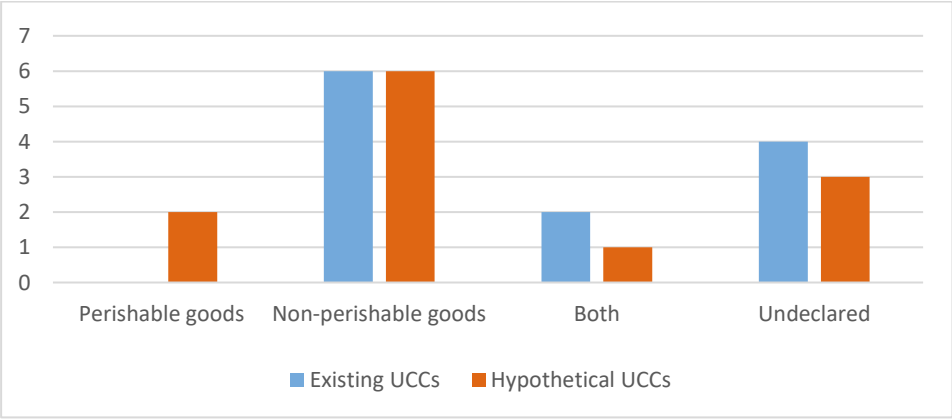


Figure 10: Types of goods handled by the studied UCC

Table 5 presents a compilation of the reviewed articles, including the number of UCCs studied in each article and the perspectives considered. The full references to the reviewed articles are provided in the reference list and are marked with asterisks. As indicated in the table, most articles focus on a single UCC, while three articles examine two UCCs, and one article examines five. Of the 17 reviewed articles, the majority address the perspective of the UCC, with the receiver's perspective also considered in several cases, whereas the shipper's perspective is seldom included. A few articles incorporate multiple perspectives, including two that consider both the UCC and the receiver, and one that includes all three actors. In terms of authorship, several articles share common first authors, and some are co-authored by the same groups of researchers. Nine articles are written by authors who do not appear as authors of the other reviewed articles.

Table 5: Compilation of reviewed articles, number of studied UCCs, and perspectives considered

	Number of studied UCCs		Perspectives considered		
Reference	**Existing**	**Hypothetical**	**Shippers**	**UCCs**	**Receivers**
Akgün et al. (2024)		1			x
Aljohani and Thompson (2021)		1		x	
Björklund et al. (2017)	5			x	
Cherrett et al. (2017)		1		x	x
Dreischerf and Buijs (2022)	2		x	x	x
Grandval et al. (2019)	1			x	x
Janjevic and Ndiaye (2017a)		1		x	
Janjevic and Ndiaye (2017b)		1		x	
Johansson and Björklund (2017)		1			x
Lin et al. (2016)		1		x	
Mangano et al. (2021)		1			x
Morganti and Gonzalez-Feliu (2015)	1			x	
Paddeu (2017)	1				x
Paddeu (2021)	1	1		x	
Paddeu et al. (2017)	1				x
Paddeu et al. (2018)	1	1			x
van Heeswijk et al. (2019)		1		x	

5. Thematic analysis

The thematic analysis is structured in line with the analytical framework, first identifying services within four categories. It then examines emphasis on five performance attributes, analysing each attribute before linking them to the identified services. Finally, it explores how the articles describe the influence of service categories on distribution system performance. Each part begins with a qualitative synthesis, followed by a quantitative overview of frequencies and distributions.

5.1 Provided services

This section identifies the specific services that UCCs are reported to provide or could potentially provide, as discussed in the reviewed articles. These services are categorised under the four service categories: delivery, warehousing, reverse logistics, and pre-retailing. The section ends with the key services being identifies.

5.1.1 Delivery

Within the category of delivery, the following 11 subcategories were formulated: consolidation, customised delivery time, fixed delivery time, off-hour delivery, perishable product delivery, same day delivery, pickup-delivery, home delivery, just-in-time delivery, on call delivery, and transfers between shops.

When *consolidation* is addressed in the articles, several reasons for its use are highlighted. According to Akgün et al. (2024), retailers may benefit by receiving fewer deliveries, thereby reducing the time required to handle incoming goods. Similarly, in two of the cases studied by Dreischerf and Buijs (2022), consolidation was associated with fewer delivery occasions. For suppliers, this meant making fewer deliveries, while for receivers, it reduced goods handling. Another reason for introducing consolidation is presented by Björklund et al. (2017), where the purpose of consolidating deliveries to municipal facilities was to reduce the number of transportation vehicles on the roads, thereby contributing to a cleaner and safer environment. In the same study, consolidation is also described as a value-added service that the UCC can provide, emphasising its role in structuring and coordinating goods.

Customised delivery time is discussed more frequently in relation to existing UCCs than to hypothetical ones, although the reasons for wanting this option do not differ between the two. In a study of an existing UCC, Paddeu (2017) reported that retailers preferred to set delivery times themselves to better organise their work and manage staff hours. Closely related to this, Akgün et al. (2024) found that retailers preferred deliveries outside peak times, when customers are present in the shop, and wanted control over which goods were delivered and when. Johansson and Björklund (2017) noted that only three of 72 interviewed retailers expressed no preference when asked about delivery times. In contrast, Paddeu et al. (2017) found that some retailers were dissatisfied with the UCC because they could not select a specific delivery time.

Fixed delivery time, in contrast, is primarily discussed in relation to hypothetical UCCs. The only mention of existing UCCs appears in Björklund et al. (2017), where fixed delivery time,

together with consolidated delivery and storage space, was described as a value-added service expected to improve logistics operations for receivers. Aljohani and Thompson (2021) also identify fixed delivery time as a value-added service.

One of the subcategories within delivery stands out due to the type of goods handled: *perishable product delivery*. The UCC studied by Morganti and Gonzalez-Feliu (2015) provides delivery of perishable products and, when requested, uses refrigerated warehouses and vehicles to maintain the cold chain. However, both Morganti and Gonzalez-Feliu (2015) and Paddeu et al. (2018) argue that the implementation and feasibility of a UCC are hampered when perishable goods, such as food, are to be handled. This is due to requirements related to cold chain maintenance, hygiene standards, and short delivery times.

The remaining subcategories are discussed less frequently and often only briefly. Grandval et al. (2019) list several value-added services provided by existing UCCs, including *same-day delivery*, *transfers between shops* of the same store brand, and *off-hour delivery*, referring to times outside shop opening hours. In Paddeu's (2017) study, off-hour delivery was also provided, as the UCC remained open at night, allowing shippers to deliver during nighttime hours. According to Janjevic and Ndiaye (2017b), off-hour deliveries could significantly enhance the attractiveness of UCCs. Paddeu (2017) also identifies *just-in-time delivery* as an additional service. Morganti and Gonzalez-Feliu (2015) note that receivers could benefit from the option of *on-call delivery*, while van Heeswijk et al. (2019) highlight *home delivery* as a value-added service for retailers. Aljohani and Thompson (2021) further identify *pickup-delivery* and pick-up offerings as additional delivery services.

5.1.2 Warehousing

Within the category of warehousing, the following three subcategories were formulated: long-term storage, short-term storage, and quality and quantity check.

Long-term storage, described by Lin et al. (2016) and Grandval et al. (2019) as a value-added service, is the most widely discussed subcategory across all service categories. Johansson and Björklund (2017) found that nearly half of the 72 interviewed retailers wanted to use long-term storage at the UCC. By providing storage services, UCCs can reduce the storage space required at retailers' premises (Akgün et al., 2024; Lin et al., 2016; Mangano et al., 2021; Morganti & Gonzalez-Feliu, 2015). Dreischerf and Buijs (2022) report that this arrangement also allows suppliers to respond to demand without operating a depot within the city. Björklund et al. (2017) provide another example, where abandoned furniture from municipal facilities is stored at the UCC rather than in individual schools or offices, with municipal staff able to check availability online.

Short-term storage is another warehousing service, identified as a value-added service by both Aljohani and Thompson (2021) and van Heeswijk et al. (2019). Akgün et al. (2024) note that it can be particularly useful during holiday seasons, when delivery volumes and frequencies increase. Similarly, Johansson and Björklund (2017) report that seven of the 72 interviewed retailers required storage space during seasonal changes and peak periods such as Christmas.

Other short-term needs can arise, for example when retailers experience a temporary lack of storage capacity (Paddeu, 2017).

Quality and quantity checks are mentioned only once in the literature, with Aljohani and Thompson (2021) identifying it as a value-added service. They describe it as an ancillary service that retailers may request to support in-store staff with goods reception.

5.1.3 Reverse logistics

Within the category of reverse logistics, the following three subcategories were formulated: recycling, waste management, and product return.

Recycling is identified as a value-added service by Grandval et al. (2019), while Paddeu (2017) and Paddeu et al. (2018) describe it as an additional service provided by the UCC. In the articles that address recycling, it is typically defined as the collection and recycling of empty packaging materials such as cardboard, plastic, or glass (Akgün et al., 2024; Grandval et al., 2019; Paddeu, 2017; Paddeu et al., 2018). Akgün et al. (2024) report that some retailers face challenges with recycling and waste management due to limited space and staff. In contrast, Johansson and Björklund (2017) note that only two of 72 interviewed retailers considered packaging materials problematic, possibly because the studied stores were located in malls, which often provide systematic on-site recycling within or near the premises.

Both Aljohani and Thompson (2021) and van Heeswijk et al. (2019) also identify *waste management* as a value-added service. According to Akgün et al. (2024), the inclusion of waste management, along with other services, could encourage retailers to adopt UCC solutions. When the UCC assumes responsibility for waste handling, it can be advantageous for retailers as it removes the need to manage it themselves (Paddeu et al., 2018).

Product return is only briefly addressed in the literature, with Aljohani and Thompson (2021) and Grandval et al. (2019) identifying it as a value-added service.

5.1.4 Pre-retailing

Within the category of pre-retailing, the following three subcategories were formulated: preparing goods for shelves and racks, attaching antitheft devices, and barcoding and tagging goods.

Akgün et al. (2024) reported that none of the 30 interviewed retailers wanted to use the UCC services of *preparing goods for shelves and racks* or *barcoding and tagging goods*. In contrast, van Heeswijk et al. (2019) highlight the activity of putting clothes on hangers as a value-added service. *Attaching anti-theft devices* is identified by Johansson and Björklund (2017) as a suitable service for a UCC, as it is something many stores may require, and centralising it could eliminate the need for each store to maintain the necessary equipment or expertise.

5.1.5 Frequencies and distribution of services

Table 6 presents the frequencies of service categories and subcategories at both the quote level and the article level. Several insights emerge from the data.

Table 6: Frequencies of service categories and subcategories in quotes and articles

	No. of quotes	% of quotes	No. of articles	% of articles
Delivery	**139**	**62.6%**	**17**	**100.0%**
Consolidation	19	8.6%	8	47.1%
Customised delivery time	17	7.7%	7	41.2%
Fixed delivery time and frequency	7	3.2%	5	29.4%
Off-hour delivery	5	2.3%	4	23.5%
Perishable product delivery	3	1.4%	2	11.8%
Same day delivery	3	1.4%	2	11.8%
Pickup-delivery	3	1.4%	1	5.9%
Home delivery	1	0.5%	1	5.9%
Just-in-time delivery	1	0.5%	1	5.9%
On call delivery	1	0.5%	1	5.9%
Transfers between shops	1	0.5%	1	5.9%
Warehousing	**33**	**14.9%**	**13**	**76.5%**
Long-term storage	23	10.4%	11	64.7%
Short-term storage	13	5.9%	9	52.9%
Quality and quantity check	2	0.9%	1	5.9%
Reverse logistics	**15**	**6.8%**	**8**	**47.1%**
Recycling	11	5.0%	6	35.3%
Waste management	7	3.2%	4	23.5%
Product return	3	1.4%	2	11.8%
Pre-retailing	**8**	**3.6%**	**5**	**29.4%**
Preparing goods for shelves and racks	2	0.9%	2	11.8%
Attaching antitheft device	2	0.9%	1	5.9%
Barcoding/tagging goods	1	0.5%	1	5.9%

The most frequently mentioned service category is *delivery*, appearing in 139 quotes (62.6%) across all 17 articles (100%). The most commonly referenced service subcategory is *long-term storage*, with 23 quotes (10.4%) across 11 articles (64.7%). Although *consolidation* and *customised delivery time* appear in more quotes than *short-term storage* (19, 17, and 13 quotes, respectively), their coverage across articles differs: *consolidation* in eight, *customised delivery time* in seven, and *short-term storage* in nine. While delivery's prominence is expected, the moderate frequency of consolidation is somewhat surprising, as it forms the core concept of a UCC. A possible explanation for its omission in more than half of the articles is that consolidation may be regarded as an inherent aspect of the UCC concept, which could lead researchers to omit explicit mention of it.

Figure 11 shows the distribution of existing and hypothetical UCCs at the quote (Figure 11a) and article (Figure 11b) levels. Overall, more quotes and articles address hypothetical UCCs in all service categories except *reverse logistics*, where existing UCCs dominate quotes and article coverage is equal. The largest difference is in *delivery*, with 52 quotes linked to existing UCCs and 86 to hypothetical UCCs.

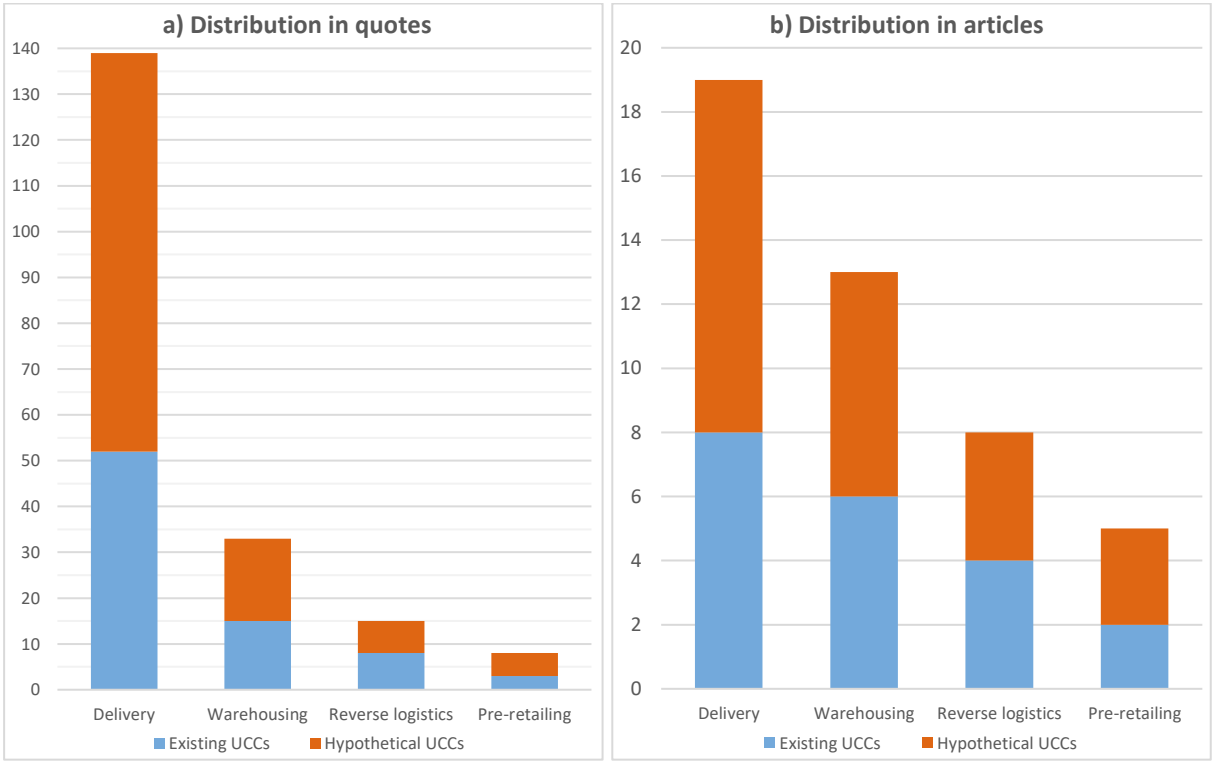


Figure 11: Distribution of existing and hypothetical UCCs across service categories in quotes and articles

Considering all quotes linked to existing (84) and hypothetical (138) UCCs, the distribution across service categories is broadly similar. *Delivery* dominates in both cases (≈62%), while *warehousing* is slightly more prominent for existing UCCs (18% vs. 13%) and *reverse logistics* also shows a higher share for existing UCCs (10% vs. 5%). *Pre-retailing* remains consistent at 3.6% for both. At the article level, *delivery* services are addressed in all articles, while *warehousing* and *reverse logistics* appear more frequently in articles on existing UCCs. *Pre-retailing* shows similar representation across both types of UCCs.

To complement the category-level analysis, the distribution is examined at the subcategory level (Figure 12) with Figure 12a representing quotes and Figure 12b articles. Hypothetical UCCs are more frequently associated with *consolidation*, *fixed delivery time, short-term storage*, and *waste management*, while existing UCCs are linked more to *customised delivery time*, *long-term storage*, and *recycling*. Some services, such as *pickup-delivery*, *home delivery*, *on-call delivery*, *quality and quantity checks*, *preparing goods for shelves*, *attaching antitheft devices*, and *barcoding/tagging*, appear only in articles on hypothetical UCCs, whereas *just-in-time delivery* and *transfers between shops* appear only in articles on existing UCCs. Pre-retailing is mentioned in general terms in three quotes from two articles but not linked to any specific service for existing UCCs.

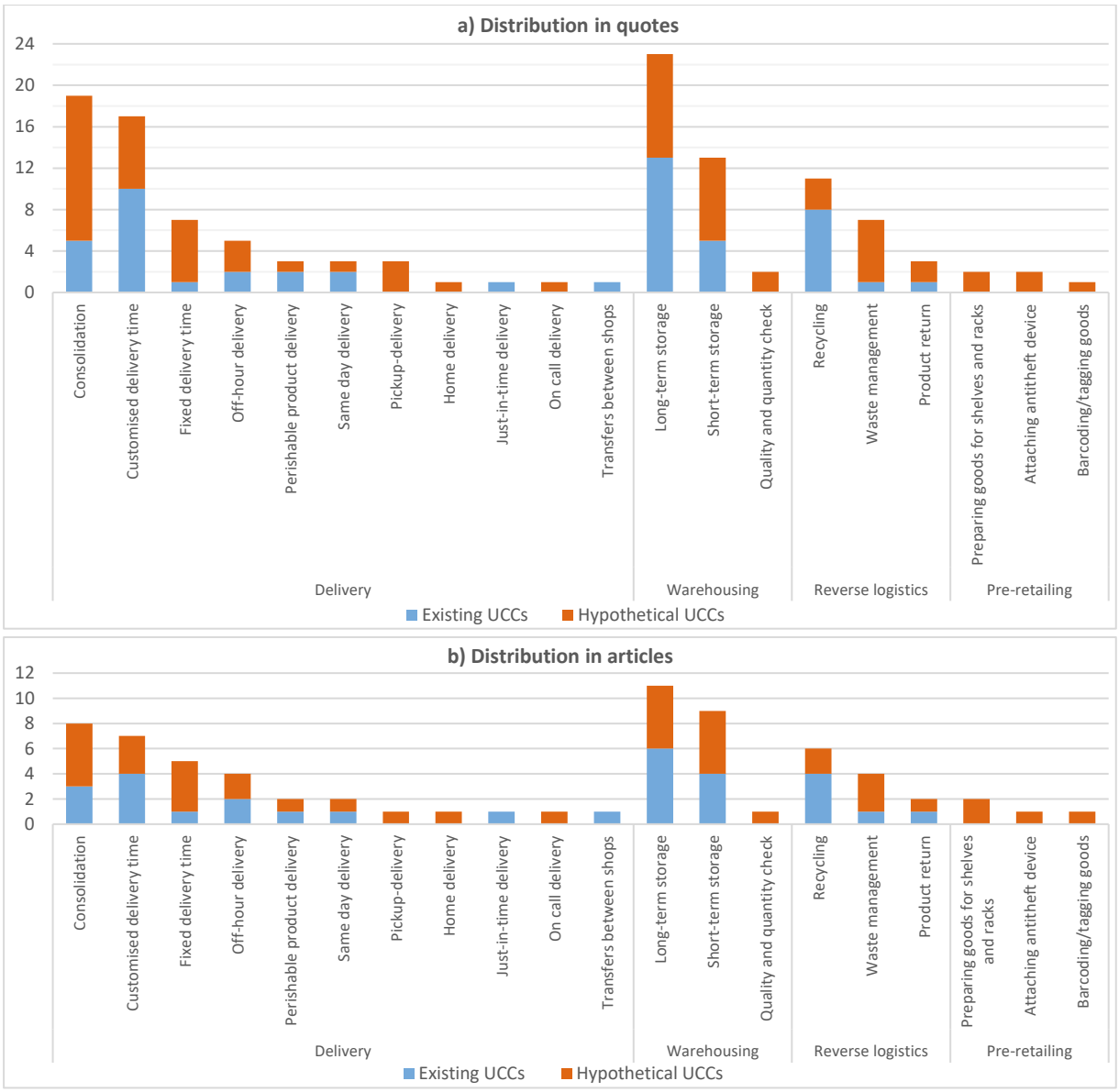


Figure 12: Distribution of existing and hypothetical UCCs across service subcategories in quotes and articles

Distinct patterns emerge when comparing services addressed in existing and hypothetical UCCs. For existing UCCs, the most frequently discussed service is *long-term storage*, followed by *customised delivery time*, *recycling, short-term storage*, and *consolidation*. In hypothetical UCCs, the order is *consolidation*, *long-term storage*, *short-term storage*, *fixed delivery time*, *customised delivery time*, and *waste management*.

Considering both qualitative descriptions and mention frequencies, five services is identified as key services that a UCC can provide: *consolidation*, *customised delivery time*, *long-term storage*, *short-term storage*, and *recycling*. *Long-term storage* is mentioned most often, followed by *consolidation*, *customised delivery time*, *short-term storage*, and *recycling*. These services consistently appear across multiple articles, highlighting their central role in UCC operations. All five key services are present in existing UCCs, though in a different order,

whereas hypothetical UCCs also feature *fixed delivery time* and *waste management*, with *recycling* appearing less prominently.

5.2 Performance

This section explores how the five predefined performance attributes are emphasised in the reviewed literature. Each attribute is analysed individually and structured into subcategories to reflect the various aspects discussed. The section then examines how frequently the attributes are addressed and linked to the specific services identified in the previous section.

5.2.1 Time

Within the time attribute the following four subcategories were formulated: receivers receipting and handling goods, delivering from UCC, and shipper delivering to the UCC.

Akgün et al. (2024) noted that by receiving fewer deliveries, the time spent by *receivers* on *receipting and handling goods* is reduced. One of the receivers interviewed by Dreischerf and Buijs (2022) emphasised that each time a delivery is made, someone, such as a security guard, must leave their post to determine who the delivery is for. Similarly, Cherrett et al. (2017) point out that consolidated deliveries can reduce interruptions to the daily routines of receiving staff. In the current system of a student hall of residence, the receptionist spent 210 minutes per day on various parcel-handling tasks, which, according to estimates, could be reduced by two hours per day through the use of consolidated deliveries via a UCC (Cherrett et al., 2017). The existing UCC studied by Paddeu et al. (2018) makes its deliveries directly to retailers' stockrooms, which saves working time for the retailers.

Paddeu (2017) noted that stores benefit from short delivery durations. However, making *deliveries from UCC* requires the goods to be handled an additional time. According to one of the suppliers interviewed by Dreischerf and Buijs (2022), this may result in deliveries taking an extra day. Similarly, Akgün et al. (2024) reported that several interviewed retailers emphasised that additional handling may lead to delays. In the modelling by Janjevic and Ndiaye (2017a), the average time spent on a delivery route serving 16.91 customers was 4.08 hours, with 12.3% of the time spent loading the vehicle, 18.6% driving, and 69.1% at customers' locations. Aljohani and Thompson (2021) emphasised that by properly training UCC staff to improve the handling process, the time required may be reduced.

Regarding *shippers delivering to the UCC*, Cherrett et al. (2017) highlight that they may also benefit from time savings by delivering all goods to a single UCC rather than to multiple locations across an urban area. Although Janjevic and Ndiaye (2017b) did not include the opportunity costs associated with time savings in their study, they argue that actors using UCC services could potentially use the time saved to serve other customers.

5.2.2 Cost

Within the cost attribute, the following four subcategories were formulated: cost to serve, integration of a UCC into the distribution system, infrastructure, and labour.

Cost to serve includes all components required to deliver a service to a customer. For consolidated deliveries, Cherrett et al. (2017) identify activities such as receiving, checking, consolidating, loading, storing, delivering, and covering staff and vehicle costs. Modelling by Janjevic and Ndiaye (2017a) shows distribution (vehicles and drivers) accounts for 63.8% of total costs, transhipment 21.1%, and general costs 15%. Thus, UCC profitability is highly sensitive to factors like receiver handling time, parcels per receiver, routes per day, and vehicle fill rate. At the same time, Johansson and Björklund (2017) note that store staff may underestimate costs borne by external actors, such as having carriers waiting due to busy staff. Consolidated delivery fees generate 73% of UCC revenue, with additional services and value-added offerings providing the remainder (Aljohani & Thompson, 2021). Akgün et al. (2024) and Paddeu et al. (2018) report that retailers may only be willing to use the UCC's services if pricing matches current arrangements, though Akgün et al. (2024) and Mangano et al. (2021) report that some may be willing to pay more for time savings, reduced hassle, and reliable deliveries. However, unclear pricing structures make it difficult to calculate costs and potential cost savings, as noted by a shipper interviewed by Dreischerf and Buijs (2022).

Integration of a UCC into the distribution system can shift costs between actors. Reductions occur primarily when UCCs fully replace another logistics step (Dreischerf & Buijs, 2022). Grandval et al. (2019) found that UCCs can reduce retailers' logistics costs; in their case, retailers paid 40% of service costs while local authorities covered 60%. Lin et al. (2016) suggest UCCs also lower costs by providing reduced storage rents and flexible deliveries, allowing storage to move from expensive retail premises to the UCC. While concerns exist among retailers about outsourcing costs (Johansson & Björklund, 2017), no evidence suggests total costs increase (Dreischerf & Buijs, 2022). Paddeu (2017) report that some retailers avoid UCCs simply due to satisfaction with current systems.

Infrastructure costs cover rent, utilities, equipment, vehicles, and fuel. Lin et al. (2016) points out that introducing a UCC adds terminal-specific costs (such as operating and rental costs at the terminal). Estimates vary: Aljohani and Thompson (2021) assign 15% to facilities (60% rent, 17% IT, 9% utilities), 13% to vehicles, and 3% to equipment; Janjevic and Ndiaye (2017a) report 4.1% rent, 7.3% vehicles, 4.8% equipment. Retailers can reduce costs by outsourcing to a UCC, centralising services like anti-theft tagging and extra storage to avoid unnecessary investments in equipment and space at each store (Johansson & Björklund, 2017), which is especially valuable in high-rent areas (Paddeu et al., 2018). Similarly, Lin et al. (2016) note that even with high terminal rents, overall infrastructure costs may drop in expensive cities by minimising in-store storage and freeing retail space. Lin et al. (2016) also report that routing deliveries through a UCC allows transfer to smaller trucks, reducing costs and transportation distance. Paddeu (2021) found that multi-drop deliveries can reduce transport costs by over 96% in a UCC scenario compared to the current system, as shown in the ex-ante evaluation.

However, vehicle costs are low compared to *labour*, which dominates overall costs; drivers and operational staff alone account for 69–75% of total costs (Aljohani & Thompson, 2021; Janjevic & Ndiaye, 2017a, b). While Janjevic and Ndiaye (2017b) used wages in Belgium in their modelling, the authors expected similar results in other high-wage countries. Aljohani and

Thompson (2021) argue that vehicle size decisions significantly influence profitability by influencing how many drivers are needed. Changes in delivery practices can reduce working hours and costs. Dreischerf and Buijs (2022) found some receivers and shippers reduced delivery frequency after introducing a UCC, aligning with Cherrett et al. (2017), who note fewer deliveries minimise interruptions for receivers. Paddeu et al. (2018) report that direct stockroom deliveries save staff time for retailers, while night-time UCC deliveries can save shipper time (Janjevic & Ndiaye, 2017b). Cherrett et al. (2017) also highlight time savings by avoiding deliveries in congested urban areas, provided they have no other customers there.

5.2.3 Quality

Within the quality attribute, the following three subcategories were formulated: punctuality, correctness, and customer satisfaction.

Morganti and Gonzalez-Feliu (2015) note that *punctual* deliveries support effective inventory management, and Akgün et al. (2024) found that retailers value having guaranteed delivery days and times via a UCC. Paddeu (2017) found that 11 of 21 UCC-using retailers reported mostly on-time deliveries, five described them as highly reliable, with staff notifying delays. However, Johansson and Björklund (2017) report that retailers not using a UCC do not express a need for more reliable schedules.

Regarding the *correctness* of deliveries, specifically in terms of shortages and damages, six out of 21 interviewed retailers felt the UCC performed better than previous arrangements, 15 rarely received damaged goods, two occasionally received damaged goods, and four never did (Paddeu, 2017). Some issues, such as wet parcels, are unavoidable in the UK (Paddeu et al., 2017). Mangano et al. (2021) highlight the importance of receiving goods in the correct quantity and quality for smaller stores, and Morganti and Gonzalez-Feliu (2015) show that limited storage makes reliable deliveries even more critical. Akgün et al. (2024) found that some retailers want to control delivery quantities to avoid receiving all goods at once. In some cases, it may be difficult to store certain goods without compromising their quality, such as food, which require short delivery times and maintained cold chains (Paddeu et al., 2018).

Customer satisfaction depends on several factors, such as service type and delivery method. UCCs can positively or negatively influence service levels, for example, changing drop-off from stockroom to doorstep may be seen as improvement or deterioration (Dreischerf & Buijs, 2022). Mangano et al. (2021) state that flexible, on-time deliveries may increase satisfaction. Paddeu et al. (2017) note dissatisfaction when retailers cannot choose delivery times, yet Paddeu et al. (2018) report high satisfaction with punctuality, correctness, and overall service provided by the UCC.

5.2.4 Productivity

Within the productivity attribute, the following four subcategories were formulated: vehicle fill rate, labour productivity, resource utilisation, and throughput.

A sensitivity analysis by Aljohani and Thompson (2021) shows that *vehicle fill rates* strongly influence UCC profitability, with a 5% increase in fill rate potentially raising net profit margin

by 4–5%. Janjevic and Ndiaye (2017a) also highlight that efficient distribution, including high fill rates, is critical for the profitability. By consolidating deliveries, fill rates may increase; for example, Paddeu (2021) estimated that for a hypothetical UCC, fill rates could raise from below 10% to 90% compared to the current system without a UCC. The same study also examined an existing UCC, which averaged a 60% fill rate but handled smaller volumes, fewer deliveries, and fewer retailers than the hypothetical case. However, introducing a UCC does not guarantee increased fill rates. Dreischerf and Buijs (2022) found that shippers may still need to enter cities for non-UCC customers, and Lin et al. (2016) note that high fill rates can sometimes be achieved without a UCC, explaining why large chain stores may not adopt UCCs.

Labour productivity is also critical Aljohani and Thompson (2021) note that handling and delivering parcels is labour-intensive and that staff training to streamline processes is essential for effective consolidation. Similarly, Janjevic and Ndiaye (2017a) stress that UCC profitability relies on efficiently organised distribution, including minimising vehicle time at customer locations. Dreischerf and Buijs (2022) found that consolidated deliveries improved logistics by reducing handling for both shippers and receivers. Johansson and Björklund (2017) noted that some retailers, in the context of hypothetical UCCs, had more staff working to handle customer service, deliveries, and in-store logistics, suggesting UCC services could reduce the need for multiple employees.

Björklund et al. (2017) identify two approaches to improving *resource utilisation*: scaling a UCC concept across cities to share systems and agreements or letting the same UCC serve several cities to form a larger logistics system, leading to increased volumes to optimise. They also note that high utilisation can be achieved on a smaller scale by using existing resources effectively. Johansson and Björklund (2017) argue that relocating store services to a UCC brings large-scale benefits, and Paddeu et al. (2018) highlight gains from sharing facilities and vehicles. Furthermore, modelling by Lin et al. (2016) indicates that the benefits of a UCC increase with customer density due to reduced logistics costs, energy consumption, and PM2.5 emissions. Janjevic and Ndiaye (2017a) found smaller service areas yield higher margins, though expanding the area by 27 times cut gross margin by only 4.9 %.

According to Aljohani and Thompson (2021), maintaining minimum daily *throughput* is essential for profitability, and Janjevic and Ndiaye (2017a) emphasise that ensuring this is necessary to avoid underutilisation, as even at low throughput at least one manager, one operator, one driver, and one vehicle with other equipment are required. Janjevic and Ndiaye (2017a) also point out that gross margin is highly sensitive to demand and throughput due to a large portion of fixed costs. In contrast, Björklund et al. (2017) found that most UCC concepts have low fixed costs, with scaling primarily influencing variable costs. These differences may reflect research design, with Janjevic and Ndiaye (2017a) modelling a hypothetical UCC and Björklund et al. (2017) studying existing ones.

5.2.5 Flexibility

Within the flexibility attribute, the following three subcategories were formulated: services provided, capacity and volume, and types of resources available.

When it comes to *services provided* by a UCC, Johansson and Björklund (2017) emphasise the need to assess service demand before implementation, as receivers' needs differ according to their specific situations. Björklund et al. (2017) add that UCCs must adapt business models and develop new services based on the current and potential customers' need. Mangano et al. (2021) note that retailers benefit from services enabling smooth deliveries, while Dreischerf and Buijs (2022) suggest UCCs can provide additional services, such as storage, not provided by suppliers or carriers, thus enhancing flexibility in the distribution system for both suppliers and receivers. However, flexibility in meeting receiver preferences may come at the expense of other attributes. Cherrett et al. (2017) argue that accommodating tailored delivery demands can be inefficient, and Paddeu et al. (2017) report one UCC charging extra for chosen delivery times. Lin et al. (2016) also show that flexibility can raise mileage, energy use, and PM2.5 emissions. Still, Johansson and Björklund (2017) emphasise that preferred delivery frequencies depend on product type and store size, leaving no universal solution.

Björklund et al. (2017) identify the ability to adjust *capacity and volume* over time as a key factor, while Janjevic and Ndiaye (2017a) recommend dimensioning resources to follow demand, for example through variable weekly working hours. Janjevic and Ndiaye (2017a) also note that seasonal and random variations in demand are harder to manage, and both Akgün et al. (2024) and Johansson and Björklund (2017) note the strong impact of holidays, which increase delivery volumes and storage needs. Paddeu (2017) adds that UCCs may also store goods temporarily when retailers lack space. At the same time, Dreischerf and Buijs (2022) stress that when introducing a UCC to the distribution system, distribution structures, including vehicles and facilities, often cannot change until current contracts are renegotiated.

Flexibility also concerns *types of resources available*, meaning the ability to use different kinds of resources based on specific needs. Lin et al. (2016) show that UCCs can split deliveries and use various vehicle combinations, which is useful in cities with size or emissions rules. Aljohani and Thompson (2021) argue that UCC operators must track receivers' needs to maintain the right vehicle mix. Available resources further shape services: Morganti and Gonzalez-Feliu (2015) note that refrigerated warehouses and vehicles allow perishable deliveries, such as food, but are limited by scarce cold-storage space.

5.2.6 Frequencies and distribution of performance

Table 7 presents the frequencies of performance attributes and subcategories at both the quote and article levels. *Cost* dominates, appearing in nearly half of the quotes (45.9%) and almost all articles (94.1%), with *cost to serve* as the most frequently mentioned subcategory. *Productivity* appears in more quotes than *flexibility*, though both are addressed in the same number of articles (12, 70.6%). *Time* and *quality* have similar quote counts, but time is covered in more articles (10 vs. 8).

Table 7: Frequencies of performance attributes and subcategories in quotes and articles

	No. of quotes	% of quotes	No. of articles	% of articles
Time	**32**	**14.4%**	**10**	**58.8%**
Receiver's receipting and handling goods	13	5.9%	7	41.2%
Delivering from UCC	12	5.4%	8	47.1%
Shipper delivering to UCC	3	1.4%	2	11.8%
Cost	**102**	**45.9%**	**16**	**94.1%**
Cost to serve	49	22.1%	13	76.5%
Integration of a UCC into the distribution system	33	14.9%	10	58.8%
Infrastructure	32	14.4%	12	70.6%
Labour	32	14.4%	10	58.8%
Quality	**32**	**14.4%**	**8**	**47.1%**
Punctuality	13	5.9%	5	29.4%
Correctness	10	4.5%	5	29.4%
Customer satisfaction	9	4.1%	5	29.4%
Productivity	**58**	**26.1%**	**12**	**70.6%**
Vehicle fill rate	27	12.2%	7	41.2%
Labour productivity	15	6.8%	7	41.2%
Resource utilisation	15	6.8%	6	35.3%
Throughput	10	4.5%	3	17.6%
Flexibility	**35**	**15.8%**	**12**	**70.6%**
Services provided	21	9.5%	10	58.8%
Capacity and volume	17	7.7%	8	47.1%
Types of resources available	4	1.8%	3	17.6%

Among non-cost subcategories, *vehicle fill rate* is mentioned most frequently in quotes (27 across seven articles), while *services provided*, *capacity and volume*, and *delivering from the UCC* appear in more articles despite fewer quotes. *Labour productivity* and *receivers' receipting and handling of goods* are addressed in seven articles, with lower quote counts.

Figure 13 shows the distribution of existing and hypothetical UCCs at quote (Figure 13a) and article level (Figure 13b). Hypothetical UCCs appear more frequently in quotes across all attributes, though the article-level pattern differs for *quality* and *productivit*y. For *quality*, quotes are mainly linked to hypothetical UCCs (19 vs. 13), while articles slightly favour existing UCCs (5 vs. 4). For *productivity*, quotes strongly emphasise hypothetical UCCs (42 vs. 16), whereas article coverage is more balanced, with some addressing both types.

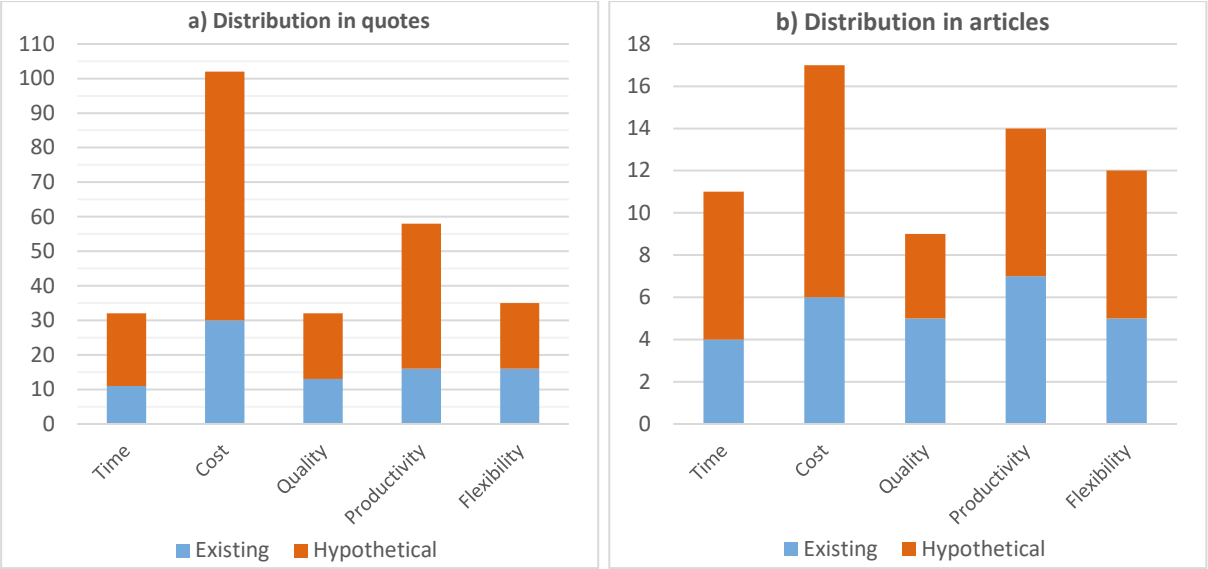


Figure 13: Distribution of existing and hypothetical UCCs across performance attributes in quotes and articles

Figure 13 also highlights differences in the distribution of quotes between existing and hypothetical UCCs for *cost* and *productivity*. While the figure shows absolute numbers, examining shares reveals that *cost* and *productivity* are proportionally more prominent in hypothetical UCCs: *cost* appears in 52% of their quotes versus 35.7% for existing UCCs, and *productivity* in 30.4% versus 19%. Differences for other attributes are smaller (*time* 2.1 percentage points, *quality* 1.7 percentage points, *flexibility* 5.2 percentage points), with existing UCCs showing slightly higher shares for *quality* and *flexibility*. This suggests that discussions of hypothetical UCCs emphasise *cost* and *productivity* more, while existing UCCs reflect a broader balance of performance attributes, possibly due to practical experience.

Figure 14 shows the distribution of existing and hypothetical UCCs across performance subcategories, with Figure 14a representing the quote level and Figure 14b the article level. In many subcategories, including *delivering from UCC*, *shipper delivering to the UCC, cost to serve*, *infrastructure*, *labour*, *punctuality*, *vehicle fill rate*, *resource utilisation*, *throughput*, and *services provided*, hypothetical UCCs are referenced more frequently. Existing UCCs, by contrast, are more often linked to *customer satisfaction*.

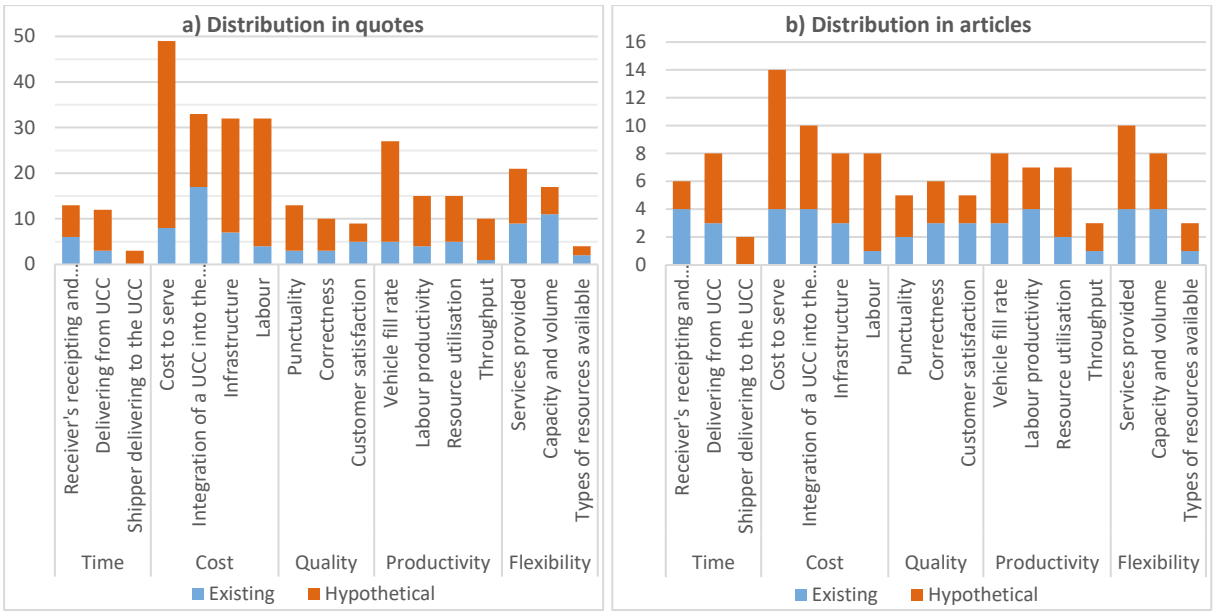


Figure 14: Distribution of existing and hypothetical UCCs within each performance subcategory in quotes and articles

Some subcategories show slight differences between quote- and article-level distributions. Three subcategories, *receiver's receipting and handling of goods*, *labour productivity*, and *correctness*, appear more frequently in quotes linked to hypothetical UCCs, though article coverage is often similar. For example, *receiver's receipting and handling of goods* has seven quotes for hypothetical UCCs versus six for existing, but more articles address existing UCCs (four vs. two). *Labour productivity* appears in 11 quotes for hypothetical UCCs and four for existing, with article coverage equal (four each). *Correctness* shows seven quotes for hypothetical UCCs and three for existing, with three articles each.

Two subcategories, *integration of the UCC into the distribution system* and *capacity and volume*, are more prominent in quotes linked to existing UCCs. *Integration of a UCC into the distribution system* has 17 quotes for existing versus 16 for hypothetical, while *capacity and volume* has 11 versus six, with article coverage equal for both subcategories.

5.2.7 Frequencies and distribution of performance linked to services

Figure 15 shows the distribution of existing and hypothetical UCCs across performance attributes and service categories at both the quote (Figure 15a) and article (Figure 15b) level. *Delivery* dominates across all attributes, reflecting its overall prominence (section 5.1.5). *Cost* is also prominent, though more articles related to existing UCCs link *delivery* to *quality* and *productivity* than to *cost*. The largest differences between existing and hypothetical UCCs occur in *delivery*: *cost* appears in 38 quotes for hypothetical versus 17 for existing UCCs, and *productivity* in 24 versus nine. At the article level, *delivery* and *productivity* are discussed in six articles for hypothetical and five for existing UCCs. In other cases, such as *warehousing* and *quality*, *warehousing* and *flexibility*, and *pre-retailing* and *quality*, the numbers are more balanced.

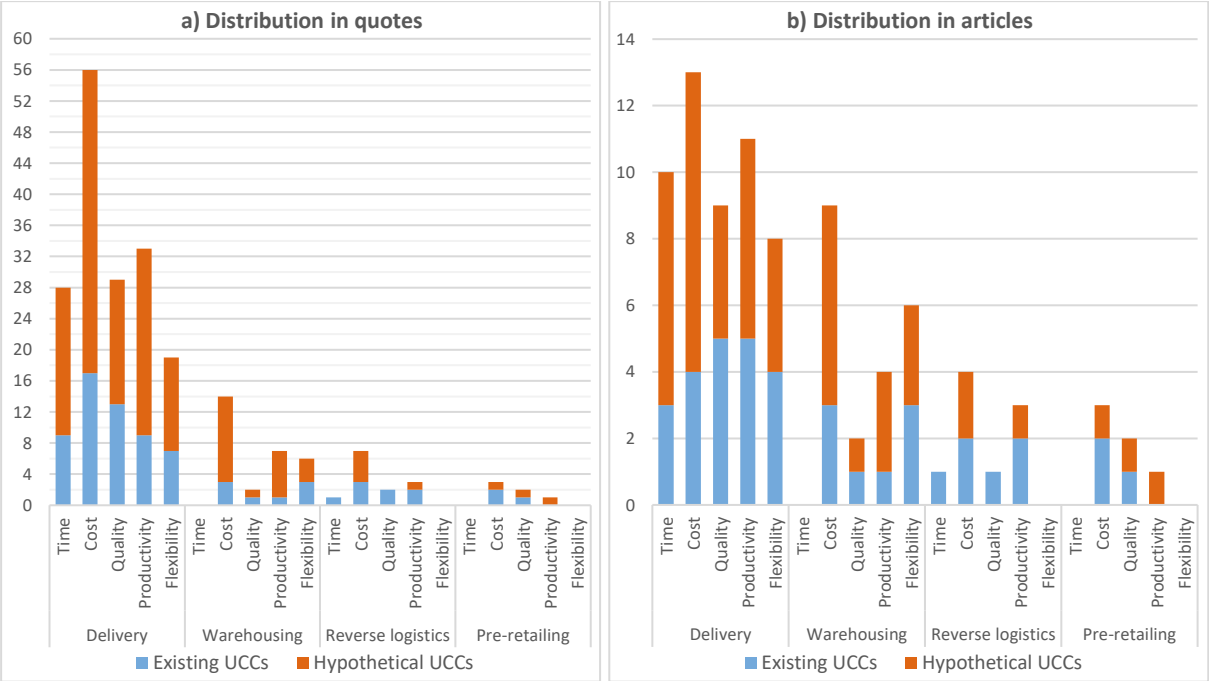


Figure 15: Distribution of existing and hypothetical UCCs within each performance attribute across service categories in quotes and articles

As noted earlier, the number of quotes and articles varies across service categories, from eight quotes in five articles for *pre-retailing* to 137 quotes in 17 articles for *delivery*. Due to this imbalance, absolute numbers alone may misrepresent how often each performance attribute is discussed per service. Figure 16 complements the analysis by showing the share of quotes and articles linking each service category to the five performance attributes at both the quote (Figure 16a) and article (Figure 16b).

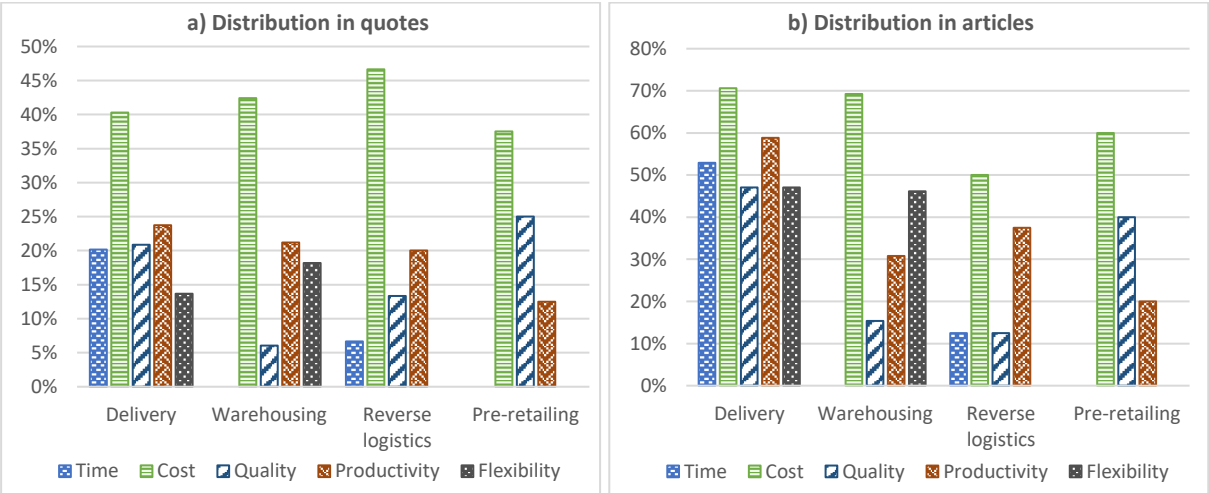


Figure 16: Share of quotes addressing both a service category and a performance attribute, relative to all quotes addressing that service category, and share of articles containing at least one such quote, relative to all articles

Cost is the most frequently discussed attribute across all service categories. Beyond *cost*, patterns differ: in *delivery*, *productivity* is second, followed by *time* and *quality*; in *warehousing*, *productivity* ranks second in quotes, but *flexibility* appears in a higher share of articles; in *reverse logistics*, *productivity* is second, followed by *time*, with a noticeable gap at the article level; *pre-retailing* is different, with *quality* second, followed by *productivity*. As noted earlier, *time* is not addressed for *warehousing* or *pre-retailing*, and *flexibility* is absent from *reverse logistics* and *pre-retailing*.

Figure 17 presents the distribution of existing and hypothetical UCCs within each performance subcategory at the quote level across service categories: delivery (Figure 17a), warehousing (Figure 17b), reverse logistics (Figure 17c), and pre-retailing (Figure 17d). In all categories except *pre-retailing*, *cost to serve* is the most frequently mentioned subcategory. Beyond this, patterns vary: *delivery* and *warehousing* cover a broader range of subcategories, while *reverse logistics* mentions only *cost to serve* and *labour productivity* for hypothetical UCCs, with a slightly wider set for existing UCCs. In *pre-retailing*, seven subcategories appear, but only once each.

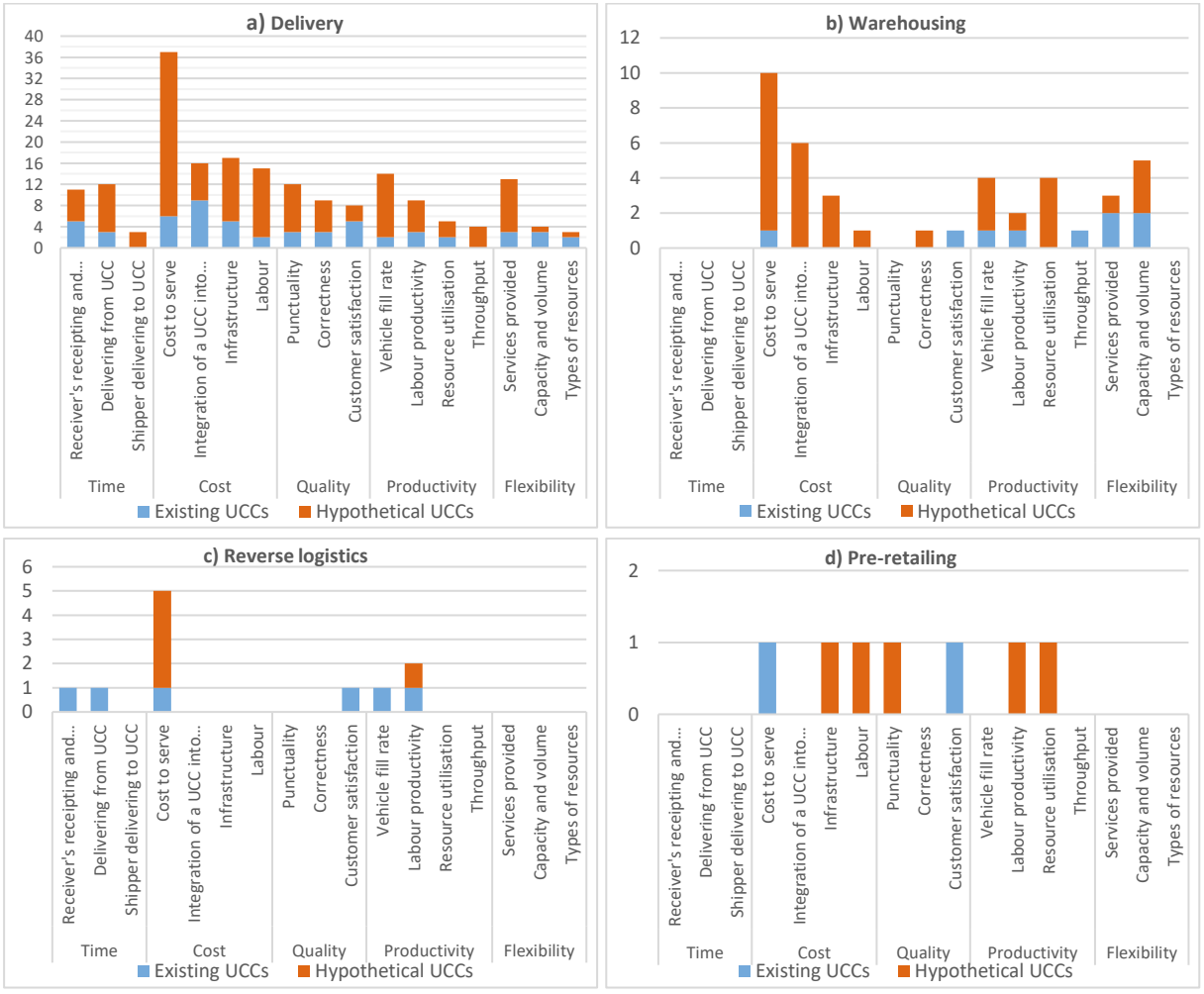


Figure 17: Distribution of existing and hypothetical UCCs within each performance subcategory across service categories in quotes

Notably, *infrastructure* and *labour* are concentrated in delivery, with 53% of *infrastructure* quotes and 47% of *labour* quotes linked to this category, while 38% of *infrastructure* and 50% of *labour* quotes are not tied to any service. This delivery-heavy distribution likely reflects the articles' focus on transport-related costs, such as vehicles, depots, and drivers.

Figure 18 presents the distribution of existing and hypothetical UCCs within each service category at the quote level across performance attributes: *time* (Figure 18a), *cost* (Figure 18b), *quality* (Figure 18c), *productivity* (Figure 18d), and *flexibility* (Figure 18e). Asterisks mark the identified key services: consolidation, customised delivery time, long-term storage, short-term storage, and recycling.



Figure 18: Distribution of existing and hypothetical UCCs within each service subcategory across performance attributes in quotes

Overall, the number of quotes linking specific service subcategories to performance attributes is low, and in more than half of the cases no connection is made. This reflects that some quotes address services collectively, for example as value-added services, without linking them to performance.

Among key services, *consolidation* is most often linked to *productivity* and *cost*, reflecting its association with operational efficiency and reduced delivery frequency. *Customised delivery time* is mainly connected to *quality*, *time*, and to some extent *flexibility*, highlighting its role in meeting receiver preferences. *Long-term* and *short-term storage* are primarily associated with *cost* and *flexibility*, suggesting support for adaptable logistics while managing storage expenses. *Recycling* is mostly linked to *cost*, with occasional mentions of *quality* and *productivity*, indicating a focus on efficiency over service responsiveness.

Customised delivery time appears more frequently in existing UCCs for *time*, *cost*, *quality*, and *productivity*, while for hypothetical UCCs it is discussed more in relation to *flexibility*. This may reflect different starting points and emphases in studies of existing versus hypothetical UCCs, which could also explain why many service subcategories are addressed predominantly in relation to one type of UCC rather than both.

5.3 The influence of services on performance

This section analyses how the reviewed articles describe the influence of UCC services on performance. The focus is on reported improvements, conditional improvements, and deteriorations, and how these effects are associated with the service categories.

5.3.1 Improvements

A major benefit of using UCCs is reducing the number of deliveries, which improves performance in several ways. Consolidating deliveries can cut time spent on transport, receipting, and handling goods for shippers, carriers, and receivers (Akgün et al., 2024; Cherrett et al., 2017; Dreischerf & Buijs, 2022; Paddeu, 2017). Cherrett et al. (2017) found that the time spent by the receiver could drop by over 40%, while Paddeu (2017) notes that fewer deliveries scheduled at convenient times help receivers organise work more productively. Reduced staff working time also lowers operational costs (Cherrett et al., 2017; Paddeu et al., 2018). Moreover, Paddeu (2017) argues that retailers can benefit from high-quality, reliable service provided from a UCC, supported with 95% of 21 retailers interviewed being very satisfied with punctuality, 90% with accuracy, and 95% with overall service. Lin et al. (2016) add that consolidation increases vehicle fill rates, which can significantly reduce transport costs (Paddeu, 2021).

Receivers may also save time and money by outsourcing activities to the UCC (Akgün et al., 2024; Johansson & Björklund, 2017). Sharing vehicles and logistics facilities reduces time and costs (Paddeu et al., 2018; Paddeu, 2021), and serving multiple receivers allows the UCC to increase productivity and achieve economies of scale (Björklund et al., 2017; Johansson & Björklund, 2017; Lin et al., 2016; Paddeu et al., 2018). This could make UCC use less costly

than a carrier, as one case in Dreischerf and Buijs (2022) showed reduced supplier transport expenses.

Located often outside central areas, UCCs can provide lower-cost storage, benefiting customers financially (Lin et al., 2016; Johansson & Björklund, 2017). Short-term storage also improves flexibility, helping suppliers handle demand spikes (Dreischerf & Buijs, 2022) and supporting receivers during holidays or limited storage periods (Akgün et al., 2024; Paddeu, 2017). Björklund et al. (2017) show that UCCs adapt business models over time to changing conditions. In one case, services like storage and inventory control for unused municipal furniture improved resource utilisation and generated additional revenue.

5.3.2 Conditional improvements

Six articles discuss various factors that, under specific conditions, indicate that using a UCC could reduce costs and increase its cost attractiveness. While almost all of these articles address hypothetical UCCs, one by Dreischerf and Buijs (2022) relates to existing UCCs. Janjevic and Ndiaye (2017a) highlight UCC throughput as one such factor, showing that adjusted gross margin rises from –92.3% at 100 packages to 28.4% at 3,000, illustrating how fixed costs are spread over larger volumes. Similarly, van Heeswijk et al. (2019) find that a UCC with high handling costs, such as rent, equipment, and staff working hours, will be unfeasible, as handling costs have a significant influence on UCC financial performance. Johansson and Björklund (2017) suggest outsourcing services needed by multiple stores reduces handling costs by avoiding duplicate equipment, staff, and space at individual stores, and increases resource utilisation by gathering the work at the UCC. Lin et al. (2016) emphasise that inexpensive storage encourages UCC use, especially when customers' own storage costs are high. However, Janjevic and Ndiaye (2017b) point out that outsourcing would introduce a service charge and suggest that overall savings depend on the size of this charge.

Modelling by Lin et al. (2016) and Janjevic and Ndiaye (2017b) shows how delivery factors influence UCC cost-effectiveness. Higher vehicle fill rates improve savings, though if direct deliveries exceed 82%, UCCs may no longer reduce costs (Lin et al., 2016). Janjevic and Ndiaye (2017b) find UCCs are more cost-effective on routes with up to seven stops, with savings dropping from 20% (one stop) to 3% (seven stops). Freight density can influence UCC attractiveness both negatively and positively: higher density reduces total distance and shipper costs, thereby decreasing the UCC's appeal, but it also lowers driving speed, which increases total service time and enhances UCC attractiveness (Janjevic & Ndiaye, 2017b). Lin et al. (2016) note that benefits rise with customer density due to economies of scale, a view supported by Dreischerf and Buijs (2022), who report that shippers anticipate cost reductions if more customers use a UCC, as it could create synergies and improve cost-efficiency.

Night-time deliveries further enhance UCC appeal; in some cases (e.g., UCCs over 15 km from depots), only night deliveries make the UCC cost-attractive (Janjevic & Ndiaye, 2017b). Even so, Dreischerf and Buijs (2022) observe that broader structural changes, such as replacing depots or transport services, renegotiating contracts, closing facilities, or reducing fleets, are often needed to realise savings.

Overall, the cost benefits of UCCs depend on certain conditions, most of which relate to productivity through resource utilisation and economies of scale. Night-time deliveries are an exception, directly saving time while indirectly improving productivity by avoiding daytime congestion.

5.3.3 Deterioration

Dreischerf and Buijs (2022) report mixed effects of UCCs on logistics costs: one shipper saw an increase, while five noted no direct change, as deliveries to other nearby customers were still required. One shipper also highlighted that a UCC serving only a single receiver limits its use for others, potentially leading to more delivery vehicles entering the urban area. Lin et al. (2016) note that at high demand, facility costs dominate logistics costs, and direct deliveries can be more efficient due to higher vehicle fill rates. The tipping point at which direct deliveries become more cost-effective is around 534 kg (1178 lbs) per store per day, which may explain why large chain stores are unlikely to use a UCC (Lin et al., 2016). Similarly, Janjevic and Ndiaye (2017b) find that deliveries of three pallets or more per stop favour direct deliveries over UCC use.

Most retailers (Akgün et al., 2024) acknowledge that a UCC adds a supply chain step, often increasing last-mile costs. Dreischerf and Buijs (2022) note that deliveries can take an extra day, while Cherrett et al. (2017) warn that consolidation may limit same-day or early next-day service. Retailers also avoid pre-retailing services due to security concerns, delay risks, and reliance on a single delivery (Akgün et al., 2024).

Delivery quality can also deteriorate. Shippers report that if the carrier changes and the new carrier drops off goods at a different location, such as the entrance instead of the stockroom, it could reduce customer satisfaction by shifting the responsibility for moving the goods to the stockroom onto the receiver (Dreischerf and Buijs, 2022). Additionally, the receiver's preferred delivery frequency, which varies depending on factors such as product type and store size (Johansson and Björklund, 2017), presents another challenge. Lin et al. (2016) highlight that providing flexible delivery frequencies to accommodate these preferences could increase vehicle miles travelled, lowering vehicle fill rates and thus reducing productivity. Paddeu et al. (2018) note instances of parcels being delivered wet, a problem the authors suggest may be unavoidable in certain climates.

5.3.4 Frequencies and distribution of the influence of services on performance

Table 8 presents how quotes describe the influence of UCC services on performance, distinguishing between improvements, conditional improvements, neutral effects, and deteriorations, along with the number of articles containing these quotes. Several findings emerge. Nearly half of all quotes are neutral, particularly linked to *cost*. Neutral quotes dominate for most attributes except *time*, where the majority (62.5%) describe improvements. *Cost* has a similar number of improvement quotes as *time*, but they represent only 18.6% of cost-related quotes due to the larger overall volume. *Quality* has the highest share of deterioration quotes, though improvements outnumber them two to one. Overall, quotes describing improvements are 4.2 times more frequent than those describing deterioration,

suggesting UCCs are more often associated with performance gains than losses. No quotes refer to a deterioration in *flexibility*. Although improvements in *productivity* and *time* appear more often in quotes than improvements in *cost*, more articles report improvements in *cost*. Notably, cost-related quotes frequently describe conditional improvements, reflecting uncertainty about whether benefits can be realised.

Table 8: Frequencies of how performance is influenced by UCC services in quotes and articles

	No. of quotes	% of quotes	No. of articles	% of articles
Improvement	**63**	**28.4%**	**13**	**76.5%**
Time	20	9.0%	7	41.2%
Cost	19	8.6%	10	58.8%
Quality	10	4.5%	3	17.6%
Productivity	23	10.4%	9	52.9%
Flexibility	11	5.0%	7	41.2%
Conditional improvement	**23**	**10.4%**	**6**	**35.3%**
Time	0	0.0%	0	0.0%
Cost	23	10.4%	6	35.3%
Quality	0	0.0%	0	0.0%
Productivity	0	0.0%	0	0.0%
Flexibility	0	0.0%	0	0.0%
Neutral	**109**	**49.1%**	**17**	**100 %**
Time	8	3.6%	5	29.4%
Cost	56	25.2%	15	88.2%
Quality	17	7.7%	7	41.2%
Productivity	31	14.0%	6	35.3%
Flexibility	24	10.8%	10	58.8%
Deterioration	**15**	**6.8%**	**6**	**35.3%**
Time	4	1.8%	3	17.6%
Cost	4	1.8%	4	23.5%
Quality	5	2.3%	3	17.6%
Productivity	4	1.8%	2	11.8%
Flexibility	0	0.0%	0	0.0%

Figure 19a-d compare existing and hypothetical UCCs. In both cases, improvements are mentioned more often than deteriorations, though patterns differ. *Cost* improvements are more often linked to existing UCCs, while conditional improvements dominate in hypothetical cases. Deterioration in *cost* also appears more frequently in hypothetical studies. This likely reflects methodological differences: studies of existing UCCs report observed outcomes, whereas hypothetical studies often rely on modelling or expectations, introducing greater uncertainty. Similarly, *quality* improvements are far more common in existing UCCs, while hypothetical cases feature more deteriorations, though the limited number of articles suggests these findings may reflect individual study focus rather than general trends. For *time*, hypothetical UCCs show more references to deterioration than existing ones, while *productivity* is the only attribute where deterioration is more often linked to existing UCCs.

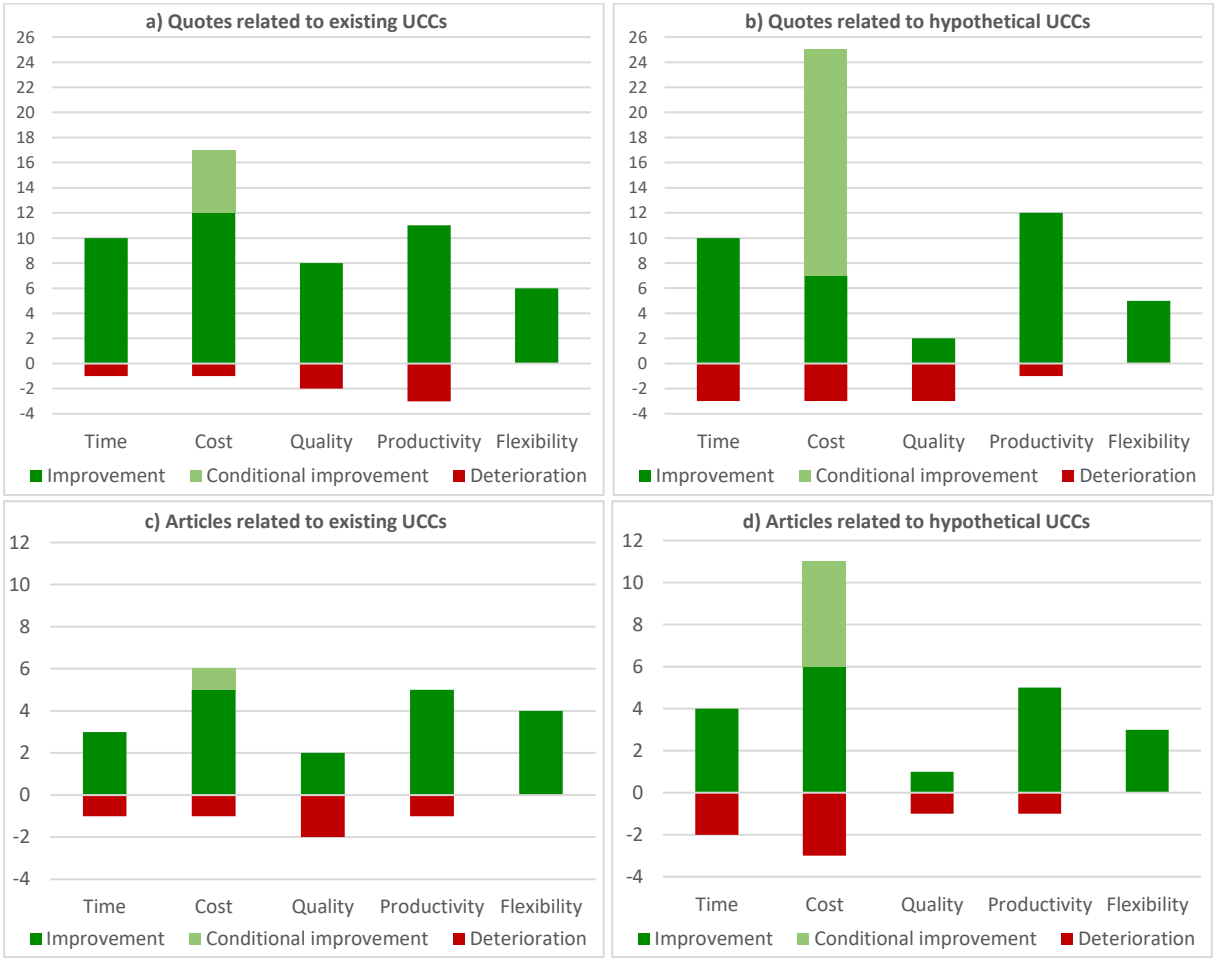


Figure 19: Distribution of improvements, conditional improvements, and deteriorations within each performance attribute for existing and hypothetical UCCs, shown in quotes and articles

Figure 20a-b further highlights differences: improvements make up a much larger share of quotes for existing UCCs, especially in *time*, *quality*, and *productivity*. For hypothetical UCCs, neutral descriptions dominate, except for *time*, which shows a more even distribution between improvement, neutral, and deterioration. This supports the interpretation that uncertainty in hypothetical studies often results in more neutral or conditional descriptions of performance. By contrast, describing performance as improvements or deteriorations may be more straightforward when studying existing UCCs.

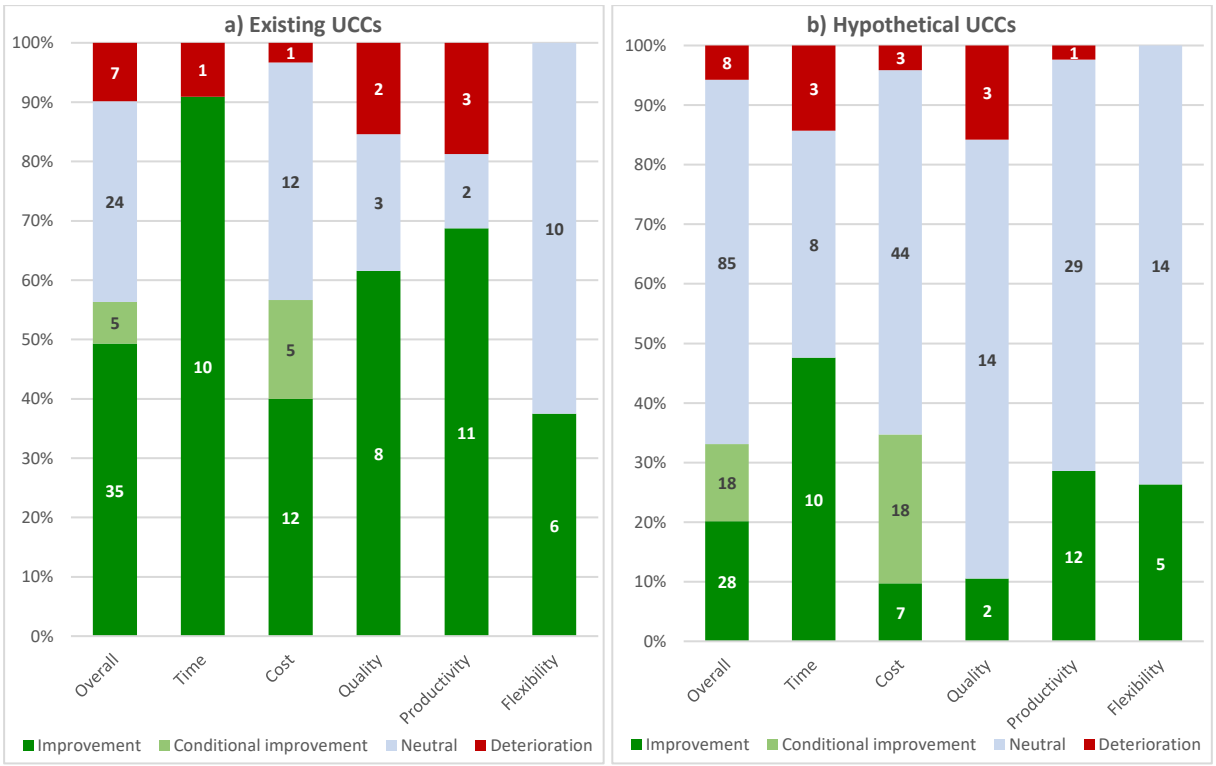


Figure 20: Share of improvements, conditional improvements, neutral effects, and deteriorations in overall performance and within each performance attribute for existing and hypothetical UCCs, shown in quotes

Figure 21 examines performance subcategories for existing UCCs (Figure 21a) and hypothetical UCCs (Figure 21b). In *time*, improvements dominate for *receiver and shipper handling goods*, but deterioration is common for *delivering form UCC*, reflecting how time-related effects vary depending on where in the distribution system the work is carried out. Cost-related subcategories differ between UCC types: existing UCC studies *emphasise integration into the distribution system*, while hypothetical ones focus on *cost to serve*. Few quotes link *cost* to deterioration.

In *quality*, improvements are spread across subcategories for existing UCCs, whereas hypothetical UCCs show limited and more negative results, particularly regarding *punctuality*. In *productivity*, *resource utilisation* has the highest number of improvement-related quotes of all attributes. *Vehicle fill rate* stands out as the only subcategory with frequent deterioration, mainly in existing UCCs. For *flexibility*, improvements are most often tied to *services provided*, followed by *capacity and volume*. This pattern may reflect the perceived added value of a UCC in enabling service differentiation and responsiveness.

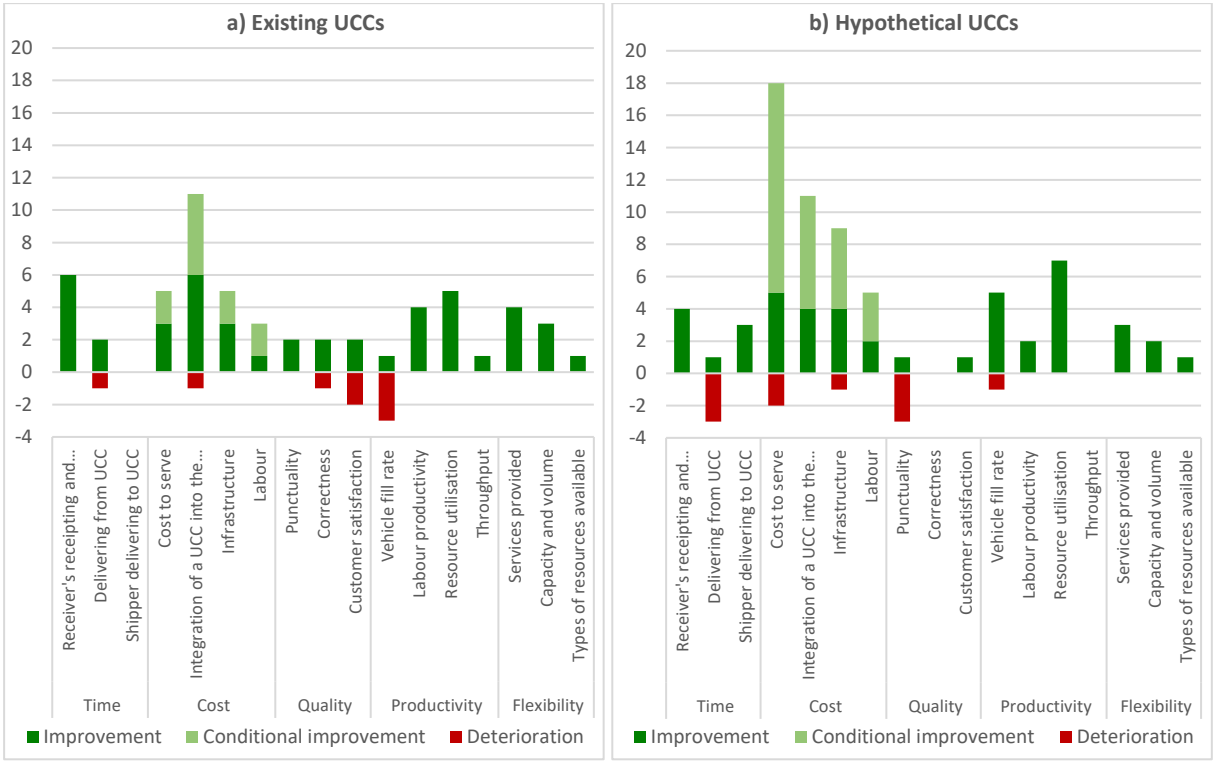


Figure 21: Distribution of improvements, conditional improvements, and deteriorations within performance subcategory for existing and hypothetical UCCs, shown in quotes

To explore how the effects on performance vary across different types of services, the analysis now turns to the distribution by service category. Figure 22 shows the distribution of improvements, conditional improvements, and deteriorations within the performance attributes across each of the four service categories. Figure 22a and Figure 22c illustrate the results for existing UCCs, where Figure 22a is based on quotes and Figure 22c on articles. Figure 22b and Figure 22d provide the corresponding results for hypothetical UCCs.

Most quotes discussing improvements, conditional improvements, or deteriorations in performance are linked to *delivery*. While quotes related to existing UCCs are more evenly distributed across service categories and performance attributes, those related to hypothetical UCCs are more concentrated. Among quotes discussing improvements, the largest difference between existing and hypothetical UCCs appears in *quality* related to *delivery*. However, the corresponding number of articles is more evenly distributed.

For existing UCCs, all quotes addressing conditional improvements concern *delivery*. In contrast, for hypothetical UCCs, conditional improvements are also discussed in relation to *warehousing* and *pre-retailing*, though *delivery* remains the most frequently addressed category. Regarding deteriorations in performance, no quotes discuss deterioration related to *reverse logistics* for either type of UCC, and none address *warehousing* or *pre-retailing* for existing UCCs. For existing UCCs, no quotes discussing deterioration in *cost* are linked to *delivery*. In contrast, for hypothetical UCCs, deterioration in *cost* related to *delivery* is the most common theme in the articles.

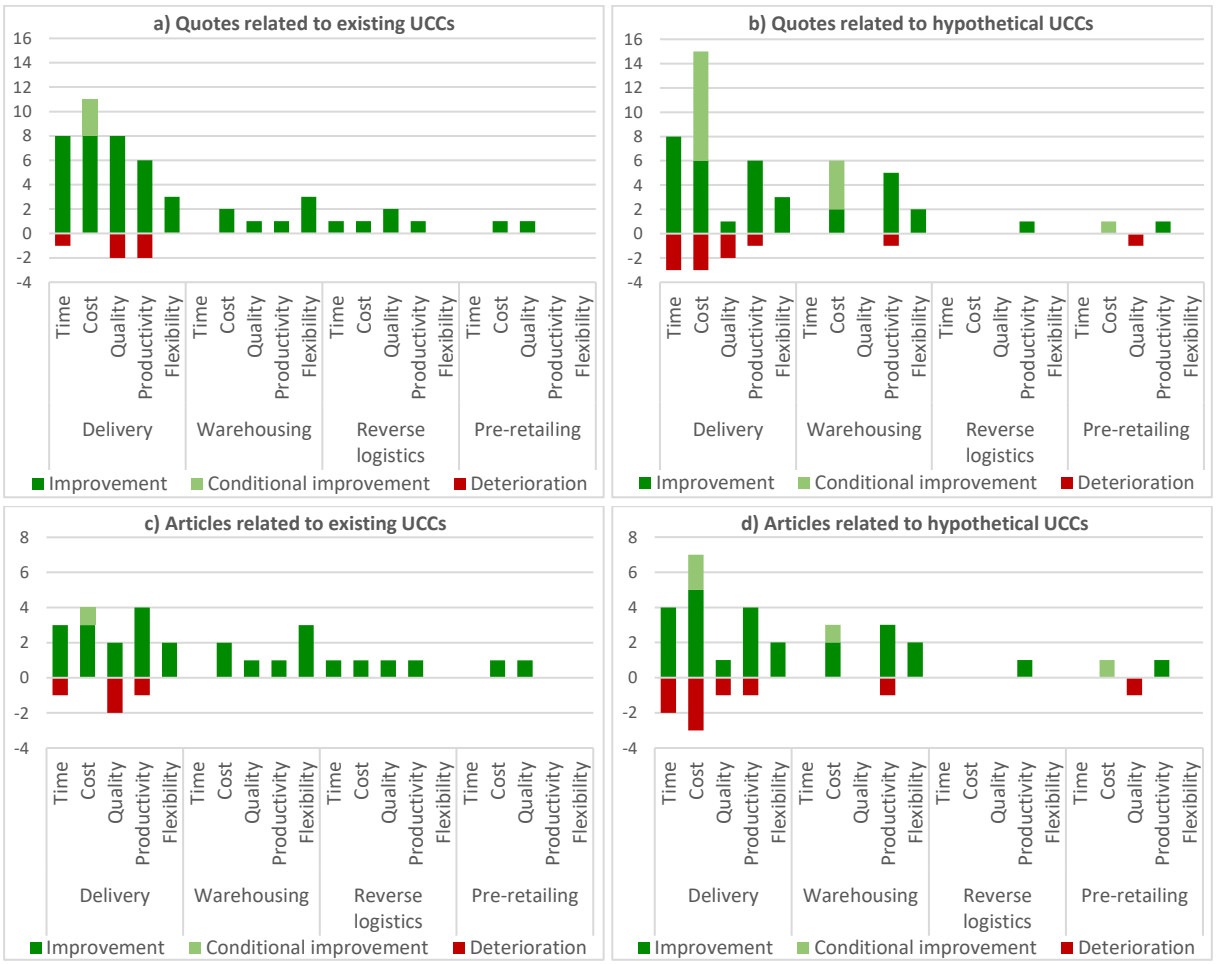


Figure 22: Distribution of improvements, conditional improvements, and deteriorations within each performance attribute across service categories for existing and hypothetical UCCs, shown in quotes and articles

Figure 23 presents the share of quotes addressing improvements, conditional improvements, neutral effects, or deteriorations within each performance attribute across service categories for existing UCCs (Figure 23a) and hypothetical UCCs (Figure 23b). Since the number of quotes varies across combinations of performance attributes and service categories, the quotes represent different shares of the total within each combination. For example, a quote addressing *cost* and *delivery* related to hypothetical UCCs accounts for only 3% of the 39 quotes in that combination, while combinations with a single quote represent 100%. Notably, the figure shows no bars for eleven combinations of performance attributes and service categories, as these were not addressed in any quotes (see Figure 15, p. 43)

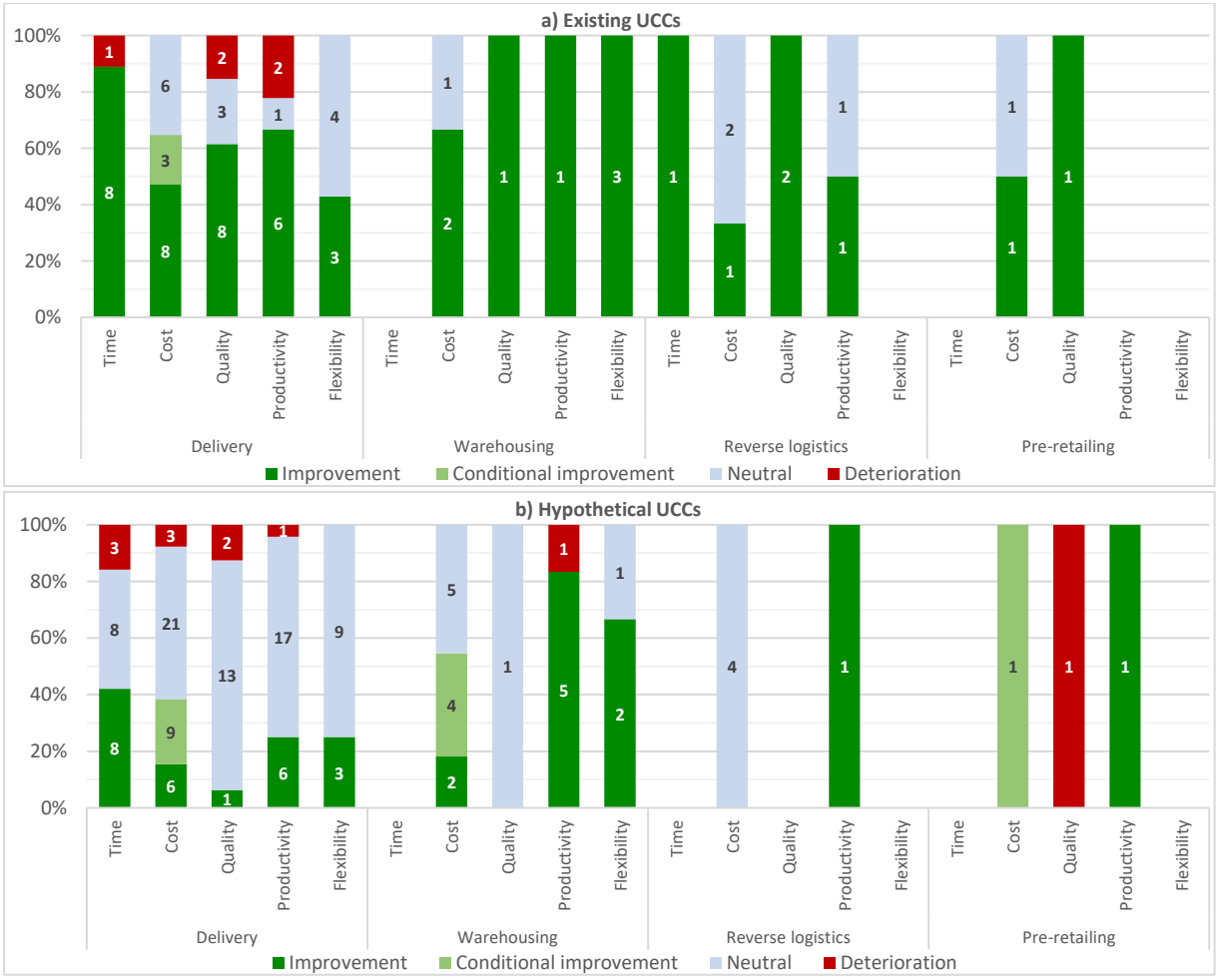


Figure 23: Share of improvements, conditional improvements, neutral effects, and deteriorations within each performance attribute across service categories for existing and hypothetical UCCs, shown in quotes

Figure 23 shows a similar pattern to earlier results: a larger share of quotes on existing UCCs describe performance improvements, while hypothetical UCCs are more often reported in neutral terms. Still, for hypothetical UCCs, improvements, conditional improvements, and deterioration together make up a greater proportion in six bars, five of which contain no neutral quotes. For existing UCCs, *warehousing* has the largest share of improvement-related quotes, though delivery has the highest number overall. For hypothetical UCCs, *pre-retailing* shows the largest share of improvement-related quotes, while *delivery* again records the most in absolute terms.

Finally, Figure 24 links performance to service subcategories for existing UCCs (Figure 24a) and hypothetical UCCs (Figure 24b). Most quotes concern key services: consolidation, customised delivery time, long-term storage, short-term storage, and recycling (marked with asterisks). *Consolidation* is the only key service associated with deterioration, specifically in *time*, though it is also linked to multiple improvements, indicating both efficiency gains and potential delays. *Customised delivery time* consistently improves *time*, *quality*, and *productivity*. *Long-term storage* shows improvements across all attributes, suggesting it supports a stable and responsive distribution system. *Short-term storage*, though less frequently

mentioned, contributes to *flexibility* and *productivity* by managing fluctuations in delivery timing and space availability. *Recycling*, discussed only for existing UCCs, is linked to improvements across several attributes.

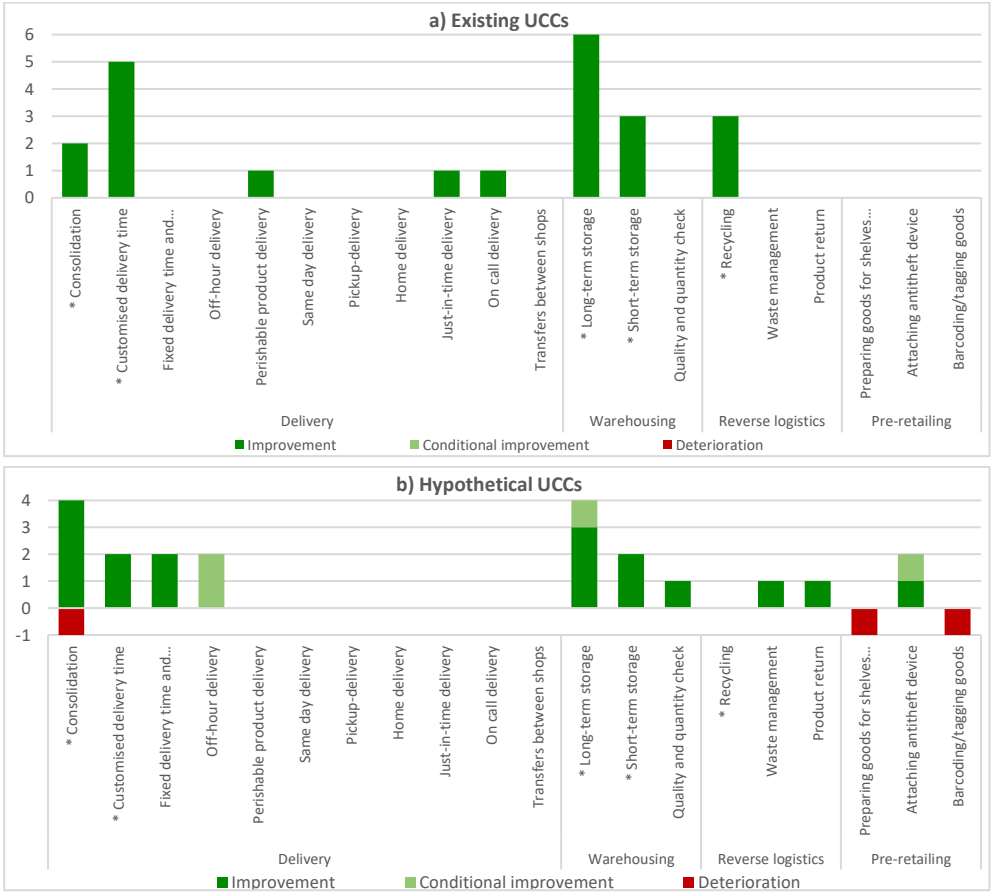


Figure 24: Distribution of improvements, conditional improvements, and deteriorations in overall performance across service subcategories for existing and hypothetical UCCs, shown in quotes

Four subcategories, *same day delivery*, *pickup-delivery*, *home delivery*, and *transfers between shops*, show no performance effects. For existing UCCs, *perishable product delivery* and *on-call delivery* improve *flexibility*, while *just-in-time delivery* improves *cost*. For hypothetical UCCs, *fixed delivery time and frequency* improve *quality* and *productivity*, and *off-hour delivery* shows conditional *cost* improvements depending on nighttime access. Improvements in *productivity* are also linked once to quality and quantity checks, *waste management*, and *product returns*.

Pre-retailing services are rarely discussed and associated with mixed performance effects, making their role less clear-cut. Their suitability may depend on local receiver needs and preferences.

6. Discussion and conclusion

This chapter presents the discussion and conclusions, fulfilling the thesis's purpose of reviewing and synthesising research on UCCs from a logistics management perspective, focusing on the services provided and their influence on distribution system performance. It discusses the key services a UCC could provide (RQ1), how performance attributes are emphasised in the UCC context (RQ2), and how these services influence distribution system performance (RQ3). The chapter then considers theoretical and managerial implications, outlines future research, and concludes with limitations.

6.1 Key services that could be provided by a UCC

Browne et al. (2005) define consolidated deliveries as a core function of UCCs, alongside the potential to provide a wide range of different value-added logistics services. This diversity is reflected in the reviewed literature, which identified 20 distinct services across four categories: delivery, warehousing, reverse logistics, and pre-retailing. Over half of these services fall within the delivery category, indicating the central role that delivery services have in UCC service provision. Notably, many delivery services are mentioned only briefly and in isolated cases, such as same-day and just-in-time delivery, which is somewhat surprising given their common association with enhancing supply chain responsiveness and meeting commercial demands. In contrast, consolidation and customised delivery time were frequently mentioned in the reviewed articles and identified as two of the five key services in this thesis. Their prominence is expected, given that consolidation is commonly described as central to the UCC concept (e.g., Allen et al., 2012; Browne et al., 2005; McDermott, 1975), and the ability to choose the delivery time can be an important operational advantage.

This thesis found that long-term storage emerges as a prominent service across all categories and is emphasised more strongly than short-term storage. This finding partly contradicts the descriptions by Allen et al. (2014) and Browne et al. (2005), who characterised warehousing services at UCCs as typically serving short-term needs, such as during holiday periods. While the finding also confirms their view of short-term storage as linked to manage seasonal peaks, the greater prominence of long-term storage suggests that the role of UCCs is broader and more varied than traditionally understood. It indicates that some receivers may require continuous access to additional storage capacity, reflecting differences in operational strategies, inventory management, or the scale and nature of goods handled. More broadly, this points to storage at UCCs not only functioning as a temporary buffer for demand fluctuations but also as a more permanent component of urban logistics systems. Nonetheless, both long-term and short-term storage were identified as key services in this thesis.

Although Allen et al. (2014), Browne et al. (2005), and Giampoldaki et al. (2023) note that UCCs can support reverse logistics, including return transport, packaging, and recycling, this is seldom reflected in the reviewed literature. This limited focus may stem from established reverse logistics systems operating independently of UCCs, as well as prevailing norms, cultural attitudes, and environmental policies. In regions with strong regulations, incentives, and environmental awareness, well-developed systems may encourage engagement.

Conversely, in areas with weaker regulations, lower awareness, and less developed or decentralised systems, participation in reverse logistics may be limited. Despite this overall limited attention, recycling was frequently mentioned and identified as a key service in this thesis.

Pre-retailing services are discussed extensively by Allen et al. (2014), Browne et al. (2005), and Giampoldaki et al. (2023), who describe a wide range of such services. However, the findings of this thesis identified only three specific pre-retailing services, with roughly half of the references referring to the category in general terms. This limited attention suggests that the provision of pre-retailing services is generally restricted in practice. It may reflect that such services are implemented selectively, depending on the receiver's needs, and are not a standard component of UCC operations.

Service provision differs between existing and hypothetical UCCs, revealing varying priorities in how these centres are conceptualised and operated. Existing UCCs emphasise services such as long-term storage and customised delivery times, addressing practical challenges related to inventory management, scheduling flexibility, and diverse receiver needs. Recycling highlights the role of reverse logistics in managing returns and supporting sustainable material flows. In contrast, hypothetical UCCs focus more on consolidation and fixed delivery times, reflecting an emphasis on transport efficiency and standardisation. References to waste management in these discussions may indicate alignment with sustainability objectives, whether by intention or as a result of operational considerations. Overall, these contrasts highlight how UCCs can be understood in different ways, shaped by context, objectives, and assumptions about their role in urban logistics.

In conclusion, key UCC services include consolidation, customised delivery time, long- and short-term storage, and recycling. Delivery services are central at the overall category level, while long-term storage is the most prominent subcategory. Consolidation remains the core function, supporting operational efficiency, and customised delivery time addresses important receiver needs. The presence of both long- and short-term storage indicates that UCCs can serve a broader range of logistical demands than previously assumed, extending beyond temporary warehousing. Recycling, as part of reverse logistics, is another key service, although other reverse logistics and pre-retailing services are less common. Differences between existing and hypothetical UCCs show that service provision varies with operational priorities: existing UCCs emphasise flexible storage and customised deliveries, whereas hypothetical UCCs prioritise consolidation and fixed delivery times. These findings highlight the diverse potential of UCC services and their implications for designing urban logistics systems.

6.2 Emphasis of performance attributes in UCC contexts

The findings of this thesis indicate that cost is the primary performance attribute in UCC studies, appearing in nearly half of all coded instances and in all but one article, with its subcategories also the most prevalent. This strong focus on cost reflects broader trends in logistics and operations management, where cost is treated as a key performance attribute (Slack and Lewis, 2024). Frameworks by Jothimani and Sarmah (2014), Slack and Lewis (2024), and Staudt et al.

(2015) all recognise cost as a central attribute, further emphasising its importance in evaluating logistics performance. Similarly, Ghaouta et al. (2023) observed cost as the most frequent attribute in their systematic review of warehouse performance, although to a lesser extent than in UCC studies. Together, these observations highlight the enduring significance of cost in assessing logistics performance, including within UCC services.

Productivity is the next most emphasised attribute, with vehicle fill rate emerging as its main subcategory. The findings show that cost and productivity were frequently addressed together, which is unsurprising given that productivity is defined as the output–input relationship (Lumsden et al., 2019). Higher productivity can be understood as achieving greater output with the same input, which is naturally associated with lower cost per unit. These attributes are also closely linked to time, particularly labour time, since reducing hours can simultaneously improve productivity and lower costs. Higher punctuality similarly enhances labour productivity through better planning and workload management, potentially reducing costs. This interconnected perspective reflects a systems view, consistent with Oskarsson et al. (2024), who argue that optimising individual parts of the supply chain is insufficient and that overall efficiency must be considered at the system level.

Time and quality were found to be the least emphasised performance attributes in the reviewed literature, in contrast to previous logistics studies, where delivery speed and punctuality are often key (Oskarsson et al., 2021; Jazairy and von Haartman, 2019; Nylander et al., 2023). Instead, the findings of this thesis suggest that attention to time is evenly divided between receivers' handling and receipting time and delivery times from the UCC itself, indicating that, although delivery time is important in broader logistics contexts, it is not yet a central focus in UCC performance evaluations. The relative underrepresentation of time may also reflect operational considerations, such as balancing delivery schedules with vehicle fill rates and productivity.

When flexibility is addressed, it is often linked to the variety of services provided and how these can assist receivers with short-term storage during periods of higher demand. This corresponds to what Slack and Lewis (2024) describe as range flexibility, which refers to the ability to adapt to different types of demands. In contrast, response flexibility, described by Slack and Lewis as the ability to swiftly address changes, is rarely reflected in the reviewed literature. This suggests that, in the context of UCCs, greater value is placed on the breadth of needs that can be met rather than on the speed of adapting service provision.

Comparing findings on existing and hypothetical UCCs, most performance attributes and subcategories are more frequently emphasised in hypothetical cases. However, integration into the distribution system, customer satisfaction, and capacity and volume appear slightly more often in studies of existing UCCs, likely reflecting differences in development stages. Hypothetical UCCs tend to focus more strongly on cost and productivity, often in connection with cost to serve, infrastructure, and labour, alongside broader implementation considerations. In contrast, existing UCCs are more frequently discussed as functioning components of a distribution system, where cost is often framed in terms of integration and how expenses are shared among actors. For instance, one study of a hypothetical UCC reported concerns about

outsourcing costs, whereas a study of an existing UCC observed no increase in total costs. Overall, discussions of existing UCCs tend to cover all five performance attributes more evenly, possibly because operational complexity requires attention to multiple aspects. Fully integrating into the distribution system appears involves not just cost or productivity, but also other attributes that together address coordination, user experiences, and customer needs.

Examining performance across UCC service categories reveals distinct patterns and interconnections. Cost dominates discussions in all categories, yet the attributes emphasised alongside it vary, reflecting the distinct nature and operational focus of each service. Delivery services illustrate this complexity, linking to all five performance attributes. Within the category, consolidation and customised delivery time serve complementary but different roles: consolidation primarily supports productivity and cost reduction, while customised delivery time relates more to service quality and flexibility. The findings show that a large share of delivery-related costs is linked to labour, particularly driver salaries, and that significant time may be spent at the receiver's location, highlighting the need for efficiency improvements at delivery points. Some of the reviewed articles suggest staff training to support this, ensuring operations are carried out efficiently and consistently, reducing delays and errors. Allowing carriers to leave deliveries at secure locations when receivers are unavailable could also improve performance, including shorter waiting times, but requires appropriate infrastructure and trust between the UCC and receivers.

Warehousing and reverse logistics show a different emphasis. Storage services, both short- and long-term, focus on cost, productivity, and flexibility, supporting demand management and storage efficiency, while their effects on time are more subtle but can still influence delivery coordination. Reverse logistics, exemplified by recycling, is more standardised and process-driven, emphasising cost and productivity while paying less attention to time or flexibility. These patterns suggest that some services are designed to address predictable operational needs, while others influence multiple aspects of performance simultaneously.

Pre-retailing services engage with fewer performance attributes, mainly cost and quality, yet potential variation in receiver requirements hints at an understated role for flexibility. Taken together, these observations show that the impact of UCC services depends not only on service type but also on operational choices and actor priorities. Recognising these nuances provides a richer understanding of the opportunities and limitations inherent in managing UCC operations.

Reflecting on UCC services, it is clear that decisions about what services to provide involve careful trade-offs. Ensuring sufficient demand and developing the business model requires attention to current and potential customers. At the same time, actors can interpret and prioritise performance attributes differently (Slack and Lewis, 2024), making trade-off assessments even more complex. While greater flexibility or customised service provisions can bring benefits, they may also increase costs or reduce productivity if resources are limited or poorly allocated. Services that require dedicated equipment, such as installing antitheft devices or maintaining refrigerated warehouses and vehicles, impose additional capacity and investment demands. Conversely, new services can enhance productivity when existing resources are underutilised, highlighting the balance between expanding capabilities and maintaining efficiency. Overall,

UCC service design and management reflect a continuous negotiation of these trade-offs, shaped by practical constraints and differing priorities.

In conclusion, cost is the most frequently emphasised performance attribute in the context of UCCs, highlighting its central role in logistics performance assessments. Productivity is the second most prominent attribute and is often considered alongside cost, as operational efficiency and cost tend to be intertwined. Time, quality, and flexibility receive less attention overall, though flexibility is addressed more often than the other two. The findings of this thesis indicate that performance attributes are not emphasised uniformly across UCC services. Rather, the nature of the service, whether focused on delivery, warehousing, reverse logistics, or pre-retailing, shapes which attributes are prioritised. Delivery services are associated with all five performance attributes, reflecting their complex and varied role within distribution systems. In contrast, warehousing and reverse logistics are more narrowly linked to cost and productivity, with flexibility also relevant to warehousing, pre-retailing services are mainly associated with cost and quality. The variation across key services further illustrates how the emphasis on specific performance attributes varies depending on the service's function, implementation, and perceived role within the broader distribution system. Moreover, differences between existing and hypothetical UCCs indicate that emerging implementations tend to prioritise cost and productivity, while established UCCs address a broader range of attributes, including integration and user experience, reflecting operational complexity. These findings suggest that while cost consistently remains central, the relative emphasis of other performance attributes is shaped by the nature of the service, its operational context, and how actors interpret and balance competing objectives.

6.3 How UCC services influence distribution system performance

A commonly mentioned benefit of UCCs is their potential to increase vehicle fill rates and reduce the number of delivery vehicles operating in urban areas by consolidating shipments before final delivery (Allen et al., 2012; Johansson, 2020; McDermott, 1975). The idea is that fewer partially loaded vehicles and more consolidated deliveries improve efficiency, reduce congestion, and lower emissions. However, this thesis shows that UCCs' practical influence is more nuanced and context dependent. While hypothetical UCCs often report improved fill rates, existing UCCs frequently show deterioration, revealing a gap between theory and practice. This may occur because shippers still serve customers not connected to the UCC, meaning delivery vehicles still enter the urban area, with lower fill rates than before, despite consolidation efforts. Moreover, having UCCs serving only a limited number of receivers may introduce additional vehicle movements. The findings further suggest UCCs tend to be most effective when deliveries are fragmented and shipment sizes are small, offering clear consolidation opportunities. In these cases, routing goods through a UCC can improve fill rates and reduce vehicle numbers. Where delivery volumes per stop are already high, potential gains are limited, and direct deliveries may remain more efficient. Therefore, the influence of UCCs on vehicle fill rates and vehicle numbers is not uniform but shaped by the demand structure and the degree of UCC adoption among receivers. These findings imply that the success of UCCs in reducing vehicle numbers and improving vehicle fill rates largely depends on how extensively and effectively they are integrated into the broader distribution system.

Cost considerations are central to how UCC services are adopted and valued. Concerns about higher costs are frequently raised, often because UCCs introduce an additional distribution step that can discourage participation (Browne et al., 2005). Yet, this thesis finds that UCCs are more often associated with cost improvements than deteriorations, even if worries about rising costs persist. Reported savings stem from reduced staff time, higher vehicle fill rates, shared use of vehicles and infrastructure, and access to lower-cost storage. These outcomes align with previous studies showing that efficiencies are achieved through better use of resources. Nylander et al. (2023), for example, highlight improvements in infrastructure and staff use, while Johansson (2020) shows that receivers can lower their total costs for receiving and handling deliveries when UCCs are used effectively. However, the effects of UCCs on cost are not always transparent, making it harder to assess whether an introduction of a UCC is reducing or increasing the total cost of the distribution system. Cost reductions often depend on shifting activities away from individual receivers, which may obscure who ultimately benefits and who absorbs residual expenses. For instance, one of the reviewed studies found that receivers may overlook costs borne by others, such as driver waiting times. This suggests that perceptions of cost are shaped not only by operational outcomes but also by one's position in the distribution system. In this sense, cost is both a measurable attribute and a contested issue, where different actors may interpret the same arrangements in divergent ways.

Although cost is more frequently described as improved than deteriorated, more than half of the discussions on cost are neutral. This suggests that while UCCs have the potential to reduce costs, the outcome is not guaranteed. The frequent neutrality might also reflect that the strong focus on cost in the UCC context may stem less from the actual influence of UCCs and more from the generally high attention cost receives in logistics and operations management, an attention that appears to have transferred to the UCC setting. Cost advantages may also depend on operational and structural conditions. High throughput and shared resources are vital for spreading fixed costs and avoiding duplication, while additional fees and variable delivery conditions, such as number of stops or fill rates, can reduce or erase savings. Night-time deliveries may also create productivity gains but require adjustments to standard practices. Overall, cost advantages appear closely tied to broader structural changes and collaboration among actors, showing that realising savings depends as much on strategic alignment and coordination as on the UCC itself.

In addition to cost and fill rates, time is another performance dimension where UCCs can offer improvements, despite concerns over adding an extra distribution step. Browne et al. (2005) suggest that although shippers and receivers may experience delays or additional costs, time savings can result from reduced congestion and more efficient delivery scheduling. This thesis supports that view but finds that time improvements are primarily linked to reduced handling and receipting time for receivers. Receiving fewer, better-timed deliveries enables improved staff planning and reduces the need for extra staff. Additionally, outsourcing certain tasks to the UCC can free up receiver time. For shippers, delivering to a single UCC rather than multiple urban points can reduce overall travel time. However, delivering via a UCC to may take longer due to the added step in the process. Viewed holistically, time lost in one stage can be offset by savings in another, highlighting the need for system-wide evaluation. This need is further

reinforced by a notable limitation in existing research: among the 17 articles reviewed, only one considers the perspectives of shippers, UCC operators, and receivers together. This suggests that most studies focus on individual actor viewpoints rather than capturing the interconnected and sometimes conflicting impacts across the entire distribution system, underscoring the complexity of assessing the overall influence of UCC services.

The performance subcategories most frequently described as improved include resource utilisation, receivers' receipting and handling of goods, integration of the UCC into the distribution system, services provided, cost to serve, infrastructure, and labour productivity. A common feature across these subcategories is that the improvements are tied to operational efficiency, such as consolidation, coordinated delivery scheduling, and shared use of infrastructure enable more streamlined processes, better resource use, and smoother coordination within the distribution system. This emphasis on efficiency is also evident when comparing existing and hypothetical UCCs. Existing UCCs are most often linked to improvements in handling, integration, resource use, labour productivity, and services, suggesting that practical experiences highlight efficiencies achieved through implemented operations. Hypothetical UCCs, by contrast, place greater emphasis on resource use, cost to serve, handling, integration, and infrastructure, reflecting expectations of how services might reshape cost structures and system design. Together, these findings suggest that while expectations emphasise cost and infrastructure benefits, actual experiences highlight operational efficiencies as the key link between UCC services and improvements across performance attributes.

The influence of UCCs on performance varies across service categories, with some offering clearer benefits than others. Warehousing shows the most consistently positive influence, particularly in productivity, flexibility, and cost, and is recognised as beneficial in both existing and hypothetical UCCs, suggesting broad agreement on its value. Delivery, however, produces more mixed outcomes: while improvements in cost, productivity, and quality are reported, deteriorations in time and quality highlight the persistent challenges of adding an extra distribution step, which is especially problematic for perishable or time-sensitive goods. For other goods, the extra delivery time may be less significant and might have to be accepted as part of the process. This aligns with earlier discussion of operational trade-offs, where longer delivery times may be tolerated in exchange for benefits such as increased vehicle fill rates and improved productivity. Given consolidation's far more frequent positive influence, it appears generally beneficial, though its suitability depends on the nature of the goods.

Reverse logistics stands out for its uniformly positive influence on productivity and quality, whereas pre-retailing shows both gains and occasional negative influence on quality. These contrasts indicate that while certain services, like warehousing and reverse logistics, are reliably seen as enhancing system performance, others, especially delivery, are more sensitive to operational conditions. This suggests that the benefits of UCCs are not evenly distributed across services but depend on how well each service aligns with broader system dynamics and implementation practices.

When discussing service provision, Allen et al. (2012) and Tsiulin et al. (2017) emphasise that incorporating value-added services can enhance the attractiveness of UCCs to potential users and support their viability, although Allen et al. (2012) caution that such services may introduce additional logistical complexity. In the reviewed literature, several services are described as value-added, including consolidation (Björklund et al., 2017), long- and short-term storage (Grandval, 2019; Lin et al., 2016; Aljohani & Thompson, 2021), and recycling (Grandval, 2019; Aljohani & Thompson, 2021), along with less frequently mentioned services such as transfers between shops (Grandval, 2019) and quality and quantity checks (Aljohani & Thompson, 2021). However, there is limited consensus on what qualifies as value-added, as the criteria for distinguishing such services are rarely defined.

Considering their influence on performance, the identified key services (long-term storage, short-term storage, customised delivery time, recycling, and consolidation) can also be understood as value-added, defined as services that deliver advantages for which customers are willing to pay (Tsiulin et al., 2017). The findings show that these services improve performance across all performance attributes, and these improvements reflect benefits that shippers and receivers would likely be willing to pay for. Long- and short-term storage primarily improve cost, productivity, quality, and flexibility; customised delivery time improves time, quality, and productivity; and recycling is solely linked to improvements in time, cost, quality, and productivity, though only in existing UCCs. Consolidation also generates improvements in time, cost, and productivity, but differs in being the only service associated with performance deterioration, due to delays caused by the additional delivery step. This variation highlights that while the key services can reasonably be seen as value-added, their contribution to UCC viability depends on how consistently they deliver advantages and under what conditions they remain attractive to users.

The findings of this thesis suggest that the influence of UCC services on performance is unlikely to be a major contributor to the challenges UCCs face in achieving viability, as highlighted in previous research (Allen et al., 2012; Mansouri et al., 2023; van Rooijen & Quak, 2014; Tsiulin et al., 2017; Johansson, 2020). Rather than indicating negative effects, the findings suggest that both shippers and receivers are, in most cases, likely to benefit from using a UCC. This contrasts with the emphasis in earlier literature on financial and operational difficulties, and points to the possibility that barriers to viability may lie in other factors, such as the need for appropriate infrastructure, strategic alignment, and effective collaboration and mutual trust among actors. Nonetheless, these findings are based on five performance attributes (time, cost, quality, productivity, and flexibility) which, while covering much of what is typically assessed in operational performance, may not capture all aspects that influence the long-term success of a UCC.

In conclusion, the services provided by UCCs generally have a positive influence on the performance of the distribution system across all performance attributes, particularly by improving operational efficiency. This thesis shows that UCC services can improve vehicle fill rates, reduce delivery vehicles in urban areas, lower costs, and improve resource utilisation, particularly when services are effectively integrated and adapted to the demand structure.

However, these benefits are context dependent and not guaranteed; challenges such as increased delivery time due to additional distribution steps or limited UCC adoption can reduce or offset gains. The key services (consolidation, customised delivery time, long-term storage, short-term storage, and recycling) each have a positive influence on performance in distinct ways: long-term storage improves cost, productivity, quality, and flexibility; customised delivery time improves time, quality, and productivity; short-term storage primarily improves flexibility and productivity; recycling improves time, cost, quality, and productivity in existing UCCs; and consolidation generally improves time, cost, and productivity, though it may occasionally cause performance deterioration due to extra delivery steps. These improvements reflect benefits that shippers and receivers would likely value, suggesting the key services can be considered value-added. Taken together, these findings suggest that both shippers and receivers are likely to benefit when UCC services are appropriately implemented, and understanding their full influence requires a system-level perspective, as the realised improvements and trade-offs are shaped by interactions across different stages, actors, and operational conditions.

6.4 Theoretical implications

This thesis contributes to UCC research by addressing two gaps identified in the literature. First, it provides a synthesis of services that can be considered value-added, namely consolidation, long- and short-term storage, customised delivery time, and recycling, and demonstrates how these services influence distribution system performance in terms of time, cost, quality, productivity, and flexibility. By identifying the specific ways in which each service improves or conditionally influences performance, the study clarifies how value-added services can enhance UCC viability and attractiveness to shippers and receivers. Second, by adopting a logistics management perspective, the thesis shifts the focus from transport-centric evaluations to a broader understanding of how delivery, warehousing, reverse logistics, and pre-retailing services interact with logistics-related performance attributes.

The findings further reinforce the theoretical relevance of a systems perspective. Improvements in one area, such as vehicle fill rates or receiver's receipting and handling of goods, are interdependent with other stages and actors in the distribution system. This shows that UCC services operate within complex operational and structural conditions, providing a more integrated and nuanced conceptual view of UCCs within urban freight distribution.

6.5 Managerial implications

This thesis has examined UCC services and their impact on distribution system performance, yielding practical takeaways. The following considerations are relevant for managers, shippers, receivers, municipal officials, and policymakers involved in planning and operating UCCs.

1. Assess the needs of existing and potential customers before establishing the UCC
Understanding receiver requirements during planning is crucial. This thesis shows that services like customised delivery times and short-term storage are highly valued for managing peak demand and temporary shortages. Identifying the most relevant services enables a UCC to prioritise investments, attract customers early, and ensure operational efficiency.

2. Choose the right location for the UCC facility

Choosing a low-rent site reduces fixed costs, and locating near dense customer areas supports shorter delivery distances, higher consolidation volumes, and improved vehicle fill rates. The site must also support intended services, such as consolidation, storage, or reverse logistics, to ensure smooth operations. For example, providing long-term storage at a low-cost facility near multiple receivers can lower handling and storage costs.

3. Continuously reassess the needs of existing and potential receivers

Service demands can change over time, so regular assessment allows the UCC to adapt. This thesis shows that warehousing and recycling services are consistently beneficial, while delivery efficiency depends on operational conditions. Continuous monitoring helps maintain customer satisfaction and ensures efficient use of labour, vehicles, and infrastructure.

4. Ensure high productivity and efficient use of existing resources

Maximising efficiency within the UCC is key for cost-effectiveness. Findings show that consolidating deliveries can improve vehicle fill rates and reduce travel time, but only if staff, vehicles, and storage are well coordinated. Streamlined processes, staff training, and technology investments can enhance handling efficiency and reduce bottlenecks, particularly during peak periods.

5. Balance the pursuit of productivity with flexibility in service provision

While standardised services enhance efficiency, flexibility is needed to meet diverse customer needs. The thesis shows that offering customised delivery times and both long- and short-term storage can increase satisfaction without necessary compromising efficiency, allowing the UCC to serve more receivers while maintaining performance gains.

6. Explore the possibility of off-hour deliveries to the UCC

Delivering outside peak hours can reduce congestion, improve vehicle fill rates, and smooth the flow of goods. Findings from existing UCCs shows that off-hour deliveries can minimise delays caused by traffic or queuing at the facility, making inbound logistics more reliable and supporting better coordination with receiver schedules.

7. Organise outbound deliveries to improve efficiency for carriers and receivers

Structuring last-mile deliveries to optimise routes and reduce frequency can improve overall system performance. For instance, consolidation at the UCC can improve overall delivery efficiency by combining deliveries, while customised delivery times improve reliability and reduce handling delays at receiver locations, benefiting both carriers and receivers.

8. Include reverse logistics services in UCC's service provision

Providing reverse logistics, such as collecting recyclable materials or returns, increases vehicle utilisation, and prevents empty return trips. This thesis shows that reverse logistics consistently improves productivity and quality for receivers, making the distribution system more balanced and cost-effective.

6.6 Limitations and further research

There are a number of limitations in the study that can be addressed in further research. First, this study is based on a SLR including only peer-reviewed journal articles, excluding grey literature such as reports, conference proceedings, and other non-peer-reviewed publications. Given that much practical knowledge on existing UCCs is published outside peer-reviewed journals, including grey literature in further research could capture a broader spectrum of operational experiences, especially from less-successful or experimental UCCs. This may reveal different patterns of how UCC services influence performance and provide a more comprehensive understanding of challenges and opportunities in UCC operations.

Second, the relatively small sample of 17 articles means that findings may be disproportionately influenced by individual studies and may not fully capture the diversity of UCC implementations. Further research could expand the evidence base through multi-case studies or cross-city comparisons to investigate whether the performance patterns identified in this thesis generalise to other contexts or reveal new operational dynamics. Collecting operational data (such as vehicle fill rates, delivery times, resource utilisation, and labour cost) alongside interviews could provide a more detailed picture of real-world performance.

Third, the SLR included research published up to June 2024, when the article search was done. Studies published after this date may include additional services or performance outcomes not reflected here. Further research could adopt longitudinal or living review approaches to track the evolution of UCC services and performance over time, ensuring that emerging practices are systematically captured.

Fourth, the reviewed articles examining existing UCCs studied operations that can be considered, to varying degrees, well-functioning and successful, meaning challenges faced by less successful or failed operations may be underrepresented. Further research could examine struggling UCCs to identify obstacles in design, service provision, or integration into the distribution system. Such research could provide insights into structural or contextual factors that influence performance and support more robust theoretical and practical guidance.

Beyond these limitations, the findings also point to broader research needs, with areas under-explored or requiring further development. First, this thesis highlights the value of a systems perspective in evaluating UCC performance, but only one reviewed article simultaneously considered the perspectives of shippers, UCC operators, and receivers. Further research should adopt comprehensive systems approaches, such as multi-actor case studies or mixed-method studies combining operational data with interviews, to capture the interconnected and sometimes conflicting effects of UCC services across the distribution system. Such research could clarify how performance attributes interact across stages and actors, providing a more accurate reflection of operational realities and challenges in urban freight logistics.

Second, another area for further research concerns how different actors perceive and value the benefits of UCC services beyond cost, such as time savings, delivery flexibility, and storage options. Surveys, interviews, or choice experiments could provide a more nuanced

understanding of stakeholder priorities and expectations. This would help identify which performance attributes are most critical from the perspectives of shippers, receivers, and UCC operators, and highlight potential trade-offs that may not be captured in existing research.

Third, despite recycling being identified as a key service in this thesis, reverse logistics services receive relatively limited attention in the reviewed literature. This reflects a gap in understanding the operational conditions and constraints under which such services can be integrated into UCCs. Further research could examine how factors such as shipment volume thresholds, infrastructure availability, or regulatory environments affect the implementation of reverse logistics. Comparative studies across different cities or countries could reveal how local contexts shape the design and feasibility of these services. Additionally, empirical studies could clarify whether UCCs are the most appropriate actors to manage return flows or if alternative logistics arrangements are more effective.

Fourth, further research could explore whether distribution systems with a UCC are better able to adjust in terms of timing, cost, and operational smoothness compared to systems without a UCC. This adjustment might be required due to changes in demand or other unexpected factors. Before-and-after studies, examining a distribution system prior to and following the introduction of a UCC, could reveal how the UCC influences operational resilience and adaptability in real-world settings. Understanding this response flexibility would provide valuable insights into the practical benefits of UCCs beyond the range of services they provide, helping to inform design and operational decisions.

References

Articles included in the literature review are marked with *

* Akgün, E. Z., Monios, J., Cowie, J. & Fonzone, A. (2024) The retailer perspective on the potential for using urban consolidation centres (UCCs). *Research in Transportation Economics*, 103, pp. 1–11. DOI: 10.1016/j.retrec.2024.101413

* Aljohani, K. & Thompson, R. G. (2021). Profitability of freight consolidation facilities: A detailed cost analysis based on theoretical modelling. *Research in Transportation Economics*, 90, pp. 1–18. DOI: 10.1016/j.retrec.2021.101122

Allen, J., Browne, M., Woodburn, A. & Leonardi, J. (2012). The role of urban consolidation centres in sustainable freight transport. *Transport Reviews*, 32(4), pp. 473–490. DOI: 10.1080/01441647.2012.688074

Allen, J., Browne, M., Woodburn, A. & Leonardi, J. (2014). A review of urban consolidation centres in the supply chain based on a case study approach. *Supply Chain Forum: An International Journal*, 15(4), pp. 100–112. DOI: 10.1080/16258312.2014.11517361

Björklund, M. (2023). *Hållbara logistiksystem*. 3rd edn., Lund: Studentlitteratur.

* Björklund, M., Abrahamsson, M. & Johansson, H. (2017). Critical factors for viable business models for urban consolidation centres. *Research in Transportation Economics*, 64, pp. 36–47. DOI: 10.1016/j.retrec.2017.09.009

Björklund, M. & Johansson, H. (2018). Urban consolidation centre: A literature review, categorisation, and a future research agenda. *International Journal of Physical Distribution & Logistics Management*, 48(8), pp. 745–764. DOI: 10.1108/IJPDLM-01-2017-0050

Booth, A., Sutton, A. & Papaioannou, D. (2017). *Systematic approaches to a successful literature review*. 2nd edn., London: Sage Publications.

Browne, M., Sweet, M., Woodburm, A. & Allen, J. (2005). *Urban freight consolidation centres* [final report]. London: University of Westminster. Available at: https://ukerc.rl.ac.uk/pdf/RR3_Urban_Freight_Consolidation_Centre_Report.pdf

* Cherrett, T., Dickinson, J., McLeod, F., Sit, J., Bailey, G. & Whittle, G. (2017). Logistics impacts of student online shopping: Evaluating delivery consolidation to halls of residence. *Transportation Research Part C: Emerging Technologies*, 78, pp. 111–128. DOI: 10.1016/j.trc.2017.02.021

CIVITAS WIKI (2015). *Making urban freight logistics more sustainable* [policy note]. Available at: https://civitas.eu/sites/default/files/civ_pol-an5_urban_web.pdf

CSCMP (n.d.). SCM definitions and glossary of terms. Available at: https://cscmp.org/CSCMP/CSCMP/Educate/SCM_Definitions_and_Glossary_of_Terms.aspx [Accessed 2024-08-12]

Dowlatshahi, S. (2000). Developing a theory of reverse logistics. *Interfaces*, 30(3), pp. 143–155. DOI: 10.1287/inte.30.3.143.11670

* Dreischerf, A. J. & Buijs, P. (2022). How urban consolidation centres affect distribution networks: An empirical investigation from the perspective of suppliers. *Case Studies on Transport Policy*, 10(1), pp. 518–528. DOI: 10.1016/j.cstp.2022.01.012

van Duin, J.H.R., Quak, H. & Muñuzuri, J. (2010). New challenges for urban consolidation centres: A case study in The Hague. *Procedia - Social and Behavioral Sciences*, 2(3), pp. 6177–6188. DOI: 10.1016/j.sbspro.2010.04.029

Durach, C. F., Kembro, J. & Weiland, A. (2017). A new paradigm for systematic literature reviews in supply chain management. *Journal of Supply Chain Management*, 53(4), pp. 67–85. DOI: 10.1111/jscm.12145

Faveto, A., Traini, E., Bruno, G. & Chiabert, P. (2024). Review-based method for evaluating key performance indicators: An application on warehouse system. *The International Journal of Advanced Manufacturing Technology*, 130, pp. 297–310. DOI: 10.1007/s00170-023-12684-4

Ghaouta, A., Ouiddad, A. & Okar, C. (2023). Measuring warehouse performance: A systematic literature review. *International Journal of Industrial and Systems Engineering*, 45 (3), pp. 321–364. DOI: 10.1504/IJISE.2022.10046645

Giampoldaki, E., Madas, M., Zeimpekis, V. & Vlachopoulou, M. (2023). A state-of-practice review of urban consolidation centres: Practical insights and future challenges. *International Journal of Logistics Research and Applications*, 26(6), pp. 732–763. DOI: 10.1080/13675567.2021.1972950

Gillström, H. & Björklund, M. (2024). Assessing the benefits of urban consolidation centres: An overview based on a systematic literature review. *Transport reviews*, 44(5), pp. 972–991. DOI: 10.1080/01441647.2024.2348639

* Grandval, S., Nimtrakool, K. & Grant, D. B. (2019). Factors of adoption governing the emergence of urban consolidation centres. *Supply Chain Forum: An International Journal*, 20(4), pp. 247-265. DOI: 10.1080/16258312.2019.1631713

* van Heeswijk, W., Larsen, R. & Larsen, A. (2019). An urban consolidation center in the city of Copenhagen: A simulation study. *International Journal of Sustainable Transportation*, 13(9), pp. 675–691. DOI: 10.1080/15568318.2018.1503380

IPCC (2018). *Global Warming of 1.5°C. An IPCC special report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission*

pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty. Genève: IPCC. Available at: https://www.ipcc.ch/sr15/

ITF (2021). *ITF transport outlook 2021*. Paris: OECD Publishing. DOI: 10.1787/16826a30-en

ITF (2023). *ITF transport outlook 2023*. Paris: OECD Publishing. DOI: 10.1787/b6cc9ad5-en

* Janjevic, M. & Ndiaye, A. (2017a). Investigating the financial viability of urban consolidation centre projects. *Research in Transportation Business and Management*, 24, pp. 101–113. DOI: 10.1016/j.rtbm.2017.05.001

* Janjevic, M. & Ndiaye, A. (2017b). Investigating the theoretical cost-relationships of urban consolidation centres for their users. *Transportation Research Part A: Policy and Practice*, 102, pp. 98–118. DOI: 10.1016/j.tra.2016.10.027

Jazairy, A., Persson, E., Brho, M., von Haartman, R. & Hilletofth, P. (2023). Drones in last-mile delivery: A systematic literature review from a logistics management perspective. *International Journal of Logistics Management*, 36(7), pp. 1–62. DOI: 10.1108/IJLM-04-2023-0149

Jazairy, A. & von Haartman, R. (2019). Analysing the institutional pressures on shippers and logistics service providers to implement green supply chain management practices. *International Journal of Logistics Research and Applications*, 23(1), pp. 1–41. DOI: 10.1080/13675567.2019.1584163

Johansson, H. (2020). *Customer benefits in city logistics: Towards viable urban consolidation centres*. Doctoral thesis. Linköping: Linköping University. http://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-164522

* Johansson, H. & Björklund, M. (2017). Urban consolidation centres: Retail stores' demands for UCC services. *International Journal of Physical Distribution & Logistics Management*, 47(7), pp. 646–662. DOI: 10.1108/IJPDLM-02-2017-0114

Jothimani, D. & Sarmah, S. P. (2014). Supply chain performance measurement for third party logistics. *Benchmarking: An International Journal*, 21(6), pp. 944–963. DOI: 10.1108/BIJ-09-2012-0064

* Lin, J., Chen, Q. & Kawamura, K. (2016). Sustainability SI: Logistics cost and environmental impact analyses of urban delivery consolidation strategies. *Networks & Spatial Economics*, 16(1), pp. 227–253. DOI: 10.1007/s11067-014-9235-9

Lumsden, K., Stefansson, G. & Woxenius, J. (2019). *Logistikens grunder*. 4th edn., Lund: Studentlitteratur.

* Mangano, G., Zenezini, G. & Cagliano, A. C. (2021). Value proposition for sustainable last-mile delivery: A retailer perspective. *Sustainability*, 13(7), pp. 1–15. DOI: 10.3390/su13073774

Mansouri, B., Sahu, S. & Ülkü, M. A. (2023). Toward greening city logistics: A systematic review on corporate governance and social responsibility in managing urban distribution centers. *Logistics*, 7(19), pp. 1–20. DOI: 10.3390/logistics7010019

Mayring, P. (2014). *Qualitative content analysis: Theoretical foundation, basic procedures and software solution*. Klagenfurt. Available at: https://nbn-resolving.org/urn:nbn:de:0168-ssoar-395173

McDermott, D. R. (1975). An alternative framework for urban goods distribution: Consolidation. *Transportation Journal*, 15(1), pp. 29–39. Available at: https://www.jstor.org/stable/20712349

* Morganti, E. & Gonzalez-Feliu, J. (2015). City logistics for perishable products: The case of the Parma's food hub. *Case Studies on Transport Policy*, 3(2), pp. 120–128. DOI: 10.1016/j.cstp.2014.08.003

Nylander, A., Blinge, M. & Laan, L.-A. (2023). Urban consolidation centers: Insights from a design thinking perspective. *Transportation Research Procedia*, 72, pp. 3979–3986. DOI: 10.1016/j.trpro.2023.11.479

Oskarsson, B., Ekdahl, B. & Aronsson, H. (2021). *Modern logistik: För ökad lönsamhet*. 5th edn., Stockholm: Liber.

Ouzzani, M., Hammady, H., Fedorowicz, Z. & Elmagarmid, A. (2016). Rayyan — a web and mobile app for systematic reviews. *Systematic Reviews*, 5, pp. 1–10. DOI: 10.1186/s13643-016-0384-4

* Paddeu, D. (2017). The Bristol-Bath urban freight consolidation centre from the perspective of its users. *Case Studies on Transport Policy*, 5(3), pp. 483–491. DOI: 10.1016/j.cstp.2017.06.001

* Paddeu, D. (2021). The Five Attribute Performance Assessment (FAPA) model to evaluate the performance of an urban consolidation centre. *Research in Transportation Economics*, 90, pp. 1–9. DOI: 10.1016/j.retrec.2021.101065

* Paddeu, D., Fancello, G. & Fadda, P. (2017). An experimental customer satisfaction index to evaluate the performance of city logistics services. *Transport*, 32(3), pp. 262–271. DOI: 10.3846/16484142.2016.1146998

* Paddeu, D., Parkhurst, G., Fancello, G., Fadda, P. & Ricci, M. (2018). Multi-stakeholder collaboration in urban freight consolidation schemes: Drivers and barriers to implementation. *Transport*, 33(4), pp. 913–929. DOI: 10.3846/transport.2018.6593

Ripple, W. J. et al. (2023). The 2023 state of the climate report: Entering uncharted territory. *BioScience*, 73, pp. 841–850. DOI: 10.1093/biosci/biad080

van Rooijen, T. & Quak, H. (2014). City logistics in the European CIVITAS initiative. *Procedia - Social and Behavioral Sciences*, 125, pp. 312–325. DOI: 10.1016/j.sbspro.2014.01.1476

Slack, N. & Lewis, M. (2024). *Operations strategy*. 7th edn., Harlow: Pearson.

Staudt, F. H., Alpan, G., Di Mascolo, M. & Taboada Rodriguez, C. M. (2015). Warehouse performance measurement: A literature review. *International Journal of Production Research*, 53(18), pp. 5524–5544. DOI: 10.1080/00207543.2015.1030466

Säfsten, K. & Gustavsson, M. (2019). *Forskningsmetodik för ingenjörer och andra problemlösare*. Lund: Studentlitteratur.

Tranfield, D., Denyer, D. & Smart, P. (2003). Towards a methodology for developing evidence-informed management knowledge by means of systematic review. *British Journal of Management*, 14(3), pp. 207–222. DOI: 10.1111/1467-8551.00375

Tsiulin, S., Hilmola, O.-P. & Goryaev, N. (2017). Barriers towards development of urban consolidation centres and their implementation: Literature review. *World Review of Intermodal Transportation Research*, 6(3), pp. 251–272. DOI: 10.1504/WRITR.2017.10007065

Vleugel, J. M. & Bal, F. (2018). Regional goods delivery: How to reduce its CO_2-, NO_X- and PM10-emissions? *International Journal of Energy Production and Management*, 3(4), pp. 338–347. DOI: 10.2495/EQ-V3-N4-338-347